

M³Exam: Benchmarking Multimodal Memory for Realistic User-Agent Interactions

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Abstract

Language agents are increasingly deployed over accumulating multimodal information, yet existing benchmarks assume a human-human form with sparse visuals and straightforward content, evaluating neither reasoning over authentic multimodal file interaction nor the interpretation of concealed user information. We therefore introduce M³Exam, a query-centric multimodal conversational QA benchmark built on realistic user-agent interaction, with multi-dimensional evaluation spanning cross-modal grounding and implicit information inference. Benchmarking MLLMs and memory systems reveals persistent gaps in cross-modal grounding, cross-session reasoning, and the efficiency cost of accumulating multimodal context. We further propose M³Proctor, a multimodal memory method that detects query modality bias and consumes raw visual sources only on demand, improving accuracy by 13% while cutting index-construction time and retrieved tokens by over 70%¹.

1 Introduction

As language agents (OpenAI, 2026a; Anthropic, 2026b; Google DeepMind, 2026) move toward real-world deployment, multimodal personal assistance (Cheng et al., 2025) has emerged as a representative scenario that demands multi-turn interactions. This shift elevates the central challenge from per-turn perception to managing a long-term, heterogeneous multimodal history. A fundamental capability for such agents is durable, well-structured *multimodal memory*: the ability to store, retrieve, and reason over a fragmented trail of text, images, and documents accumulated across multiple sessions (Zhang et al., 2025). Because multi-session conversations are the primary medium through which this rich memory is naturally formed and utilized, they serve as the ideal testbed for evaluating an agent’s long-term multimodal capabilities.

¹Code and data are released at <https://anonymous.4open.science/r/M-3-Exam-128D>.

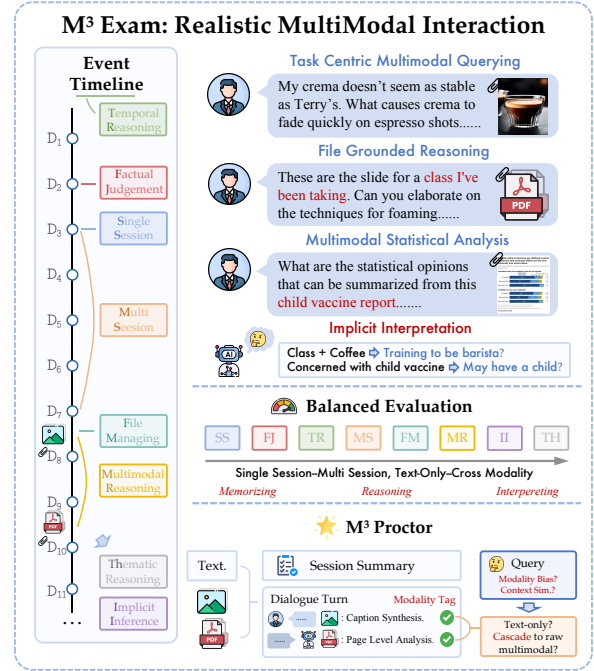


Figure 1: Overview of M³Exam.

As illustrated in Figure 1, realistic multimodal memory unfolds organically over an extended timeline: a user might share espresso photos in one session, upload class slides in another, and later analyze a child vaccine report. Answering such queries mirrors how humans recall and reason over past information: one must *memorize* what was encountered, *reason* over it to connect disparate pieces, and *interpret* information that is implied but never stated outright (Wu et al., 2025). A faithful multimodal memory benchmark must therefore capture three corresponding dimensions of difficulty. First, *content complexity* requires agents to persistently track and manage heterogeneous artifacts (e.g., dialogues, images, PDFs) within a growing, multi-session history. Second, *reasoning complexity* for cross-modal inference, like connecting a coffee’s visual state to a past PDF. Finally, *intent complexity* to deduce unstated contexts, such as inferring the user is a barista or parent from history.

Table 1: Comparison of M³Exam with representative conversational-memory benchmarks. **A. Round** = average user–assistant rounds per session; **A. File** = average images and document files per session. **MR** = multimodal reasoning over visual evidence; **SA** = statistical analysis of multimodal graphs; **FM** = file management and retrieval; **FR** = file reasoning over document content. **II** = implicit information inference; **TH** = domain-specific thematic reasoning. ✓ = supported; ⚡ = partial / restricted; ✗ = not supported; ~ = N/A.

Benchmark	Memorizing		Reasoning				Interpreting	
	A. Round	A. File	MR	SA	FM	FR	II	TH
LongMemEval (Wu et al., 2025)	5.2	~	✗	✗	✗	✗	✗	✗
MemoryArena (He et al., 2026)	6.9	~	✗	✗	✗	✗	✗	✗
MemoryAgentBench (Hu et al., 2025)	9.6	~	✗	✗	✗	✗	⚡	⚡
LoCoMo (Maharana et al., 2024)	10.8	3.4	⚡	✗	✗	✗	✗	✗
MMDialog (Feng et al., 2023)	4.6	2.6	⚡	✗	⚡	✗	✗	✗
MMRC (Xue et al., 2025)	12.9	2.9	✓	✗	✓	✗	✗	✗
Mem-Gallery (Bei et al., 2026)	16.5	4.2	✓	✗	✓	✗	✗	✗
M³Exam (Ours)	12.7	7.5	✓	✓	✓	✓	✓	✓

Existing benchmarks, however, fall short on these evaluation dimensions (Table 1). Early long-term conversational-memory suites are text-only and omit visual modalities altogether (Wu et al., 2025; He et al., 2026); subsequent multimodal benchmarks add images but treat them as static snapshots and ignore document-level artifacts (Maharana et al., 2024; Bei et al., 2026), while document-QA benchmarks probe rich multimodal reasoning but restrict it to a single document within a single turn (Tanaka et al., 2023; Ma et al., 2024). As Table 1 shows, no prior benchmark jointly stresses realistic multimodal memorizing, reasoning, and interpreting, while nearly all assume user information is fully stated, overlooking the *implicit-inference* regime that requires indirect context interpretation.

To close this gap, we introduce M³Exam, a query-centric multimodal conversational memory benchmark specifically targeting realistic multimodal user–agent interaction. M³Exam comprises 239 realistic multi-session conversations spanning 15 persona scenarios (3,025 rounds with 1,799 multimodal artifacts) paired with 5,150 evaluation questions. Given a query, an agent must analyze the accumulating multimodal history and retrieve the relevant dialogue sessions or multimodal artifacts to answer it. These demands are reflected in a reasoning-heavy, balanced question bank that, beyond standard retrieval and multi-hop questions, adds two types absent from prior work: Thematic Reasoning (TH), which requires domain knowledge implicit in the user’s context, and Implicit Inference (II), whose answer hinges on information the history implies but never states outright. Benchmarking the latest Multimodal Large Language Models (MLLMs) and frontier agent memory systems on M³Exam shows they struggle with cross-

modal reasoning and implicit interpretation.

We further turn our findings into a method. Experiments show that existing multimodal memory systems are modality-agnostic. They retrieve and inject raw visual sources indiscriminately, ignoring whether a question actually needs them, which both buries the decisive evidence among irrelevant visuals and inflates the per-query token budget. We therefore propose M³Proctor, a modality-aware memory method that detects *modality bias* within queries and consumes raw visual sources only on demand via (1) bias detection, (2) modality-aware re-ranking and (3) a cost-aware cascade that escalates to raw visuals only when text proves insufficient. On M³Exam, M³Proctor improves accuracy by over 13% with 70% less tokens while cutting index-construction time by 80% over existing systems. Overall, our main contributions are threefold:

- *Proposed Novelty Benchmark.* We introduce M³Exam, a realistic multimodal memory benchmark spanning 15 persona scenarios of 239 multi-session conversations with rich cross-modality reasoning and implicit interpretation evaluations.
- *Benchmarking & Findings.* Extensive benchmarking and in-depth analysis are conducted across latest multimodal models and agent memory systems, uncovering their limitations in visual grounding, complex implicit reasoning, and the efficiency of accumulating multimodal contexts.
- *Proposed Novelty Baseline.* We propose a novel M³Proctor, a modality-aware multimodal memory method that detects query modality bias and enables modality cascade to consume raw visual sources only on demand. M³Proctor improves accuracy by 13% with significantly less cost.

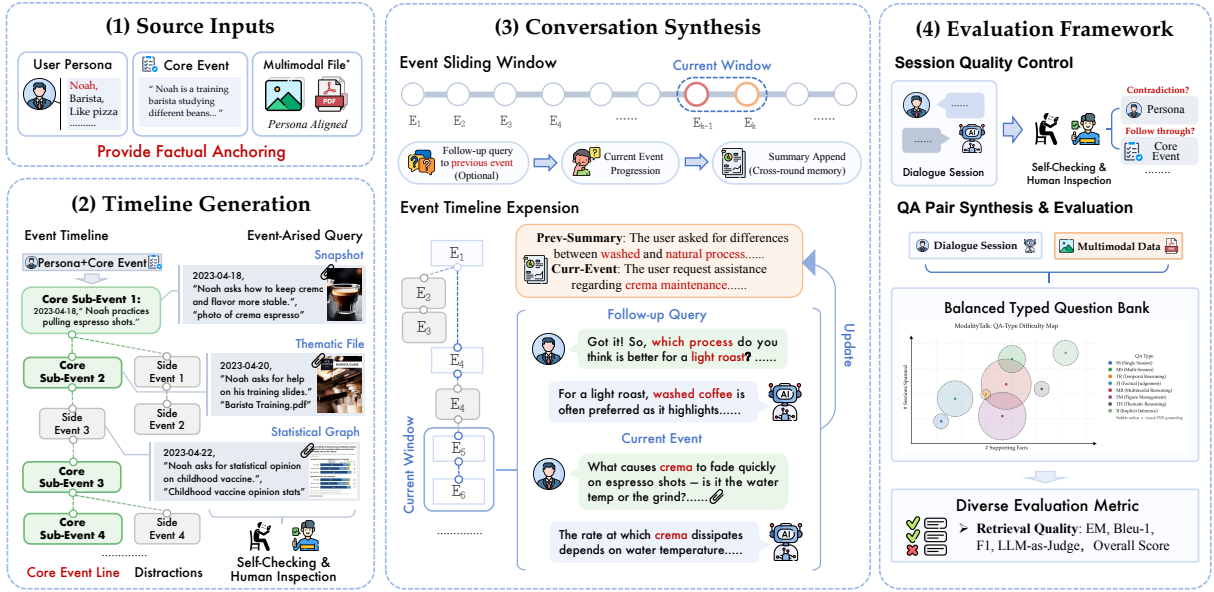


Figure 2: Overall pipeline of M³Exam, designed to evaluate multimodal memory ability in realistic scenarios.

2 M³Exam: Agent Benchmark

This section describes M³Exam in detail (Figure 2). We first formalize the task (§2.1), then describe how each instance is built from the source inputs through timeline generation, multi-turn conversation synthesis, and question-bank construction (§2.2), with quality control applied at every stage. Finally, we review the resulting benchmark statistics (§2.3).

2.1 Task Formulation

M³Exam casts the agent as a respondent over a multi-modal memory accumulated from a user’s past conversations. Each instance pairs the resulting multi-session conversation $\mathcal{D} = (D_1, \dots, D_L)$ with a query x . The sessions are ordered in time and interleave dialogue turns with attached images and document pages, forming a history $\mathcal{H}$. Given $\mathcal{H}$ and x , the agent produces an answer $\hat{a} = f(x, \mathcal{H})$, scored against the gold answer a_0 . Each query is annotated with the supporting-fact rounds $\mathcal{F}_q$ it depends on, which may span multiple sessions and modalities. Thus, answering requires *retrieving* the right evidence from $\mathcal{H}$, *grounding* it across text, images, and documents, and *interpreting* the implicit information required by the question.

2.2 Benchmark Construction

Source Inputs. Each instance is synthesized from a triple $(\mathcal{P}, \mathcal{C}, \mathcal{F})$: *Persona* $\mathcal{P}$, a short biography (name, profession, interests) that anchors and steers generation; *Core event* $\mathcal{C}$, a one-paragraph narrative arc the persona pursues over time; and a *multimodal*

file pool $\mathcal{F}$ of charts, photographs, and PDF documents, provided locally or retrieved by keyword. Construction proceeds in the following three stages, each followed by a quality-control inspection.

Timeline Construction. Conditioned on $(\mathcal{P}, \mathcal{C})$, a text large language model (LLM) generates a chronologically ordered *core-event line* of N events $\{e_1, \dots, e_N\}$, each a record $e_i = (\text{IDX}_i, \text{DESC}_i, \text{QUERY}_i, \text{TIME}_i, \text{KW}_i)$ that pairs a life event with the visual-grounded query the persona will raise, a strictly increasing date, and an image-search keyword. To mirror how real users drift off-topic, we additionally generate M *distractor side events* from domains orthogonal to $\mathcal{C}$ (errands, hobbies, casual chatter), which are merged into the core line by date and renumbered; these stress evidence retrieval amid noise and supply hard negatives for the find-matching and implicit-inference questions. An LLM self-check then audits persona violations, core-event drift, and internal contradictions, repairing only flagged events, before a human spot-check clears the timeline for synthesis.

Multi-Turn Conversation Synthesis. The timeline is extended into a conversation session by walking an *event sliding window* that advances one event per round: round k exposes a δ -event window $\mathcal{W}_k = (e_{p_k - \delta}, e_{p_k})$ whose trailing event lets the user simulator issue an optional follow-up while the leading event drives the current query over its freshly attached images, delivered cumulatively so the assistant at round k sees all images up to k . To

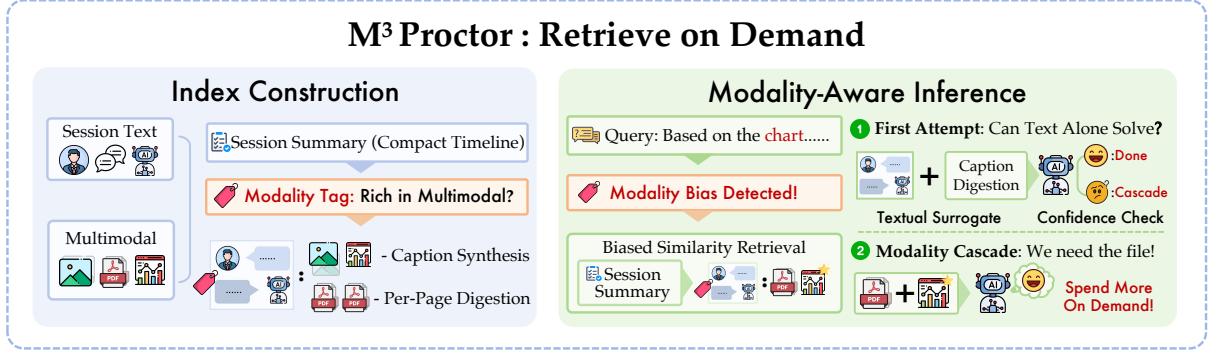


Figure 3: Overview of M³Proctor, designed to detect modality bias within queries and retrieve only on demand.

carry memory across long horizon, each exchange is condensed into a key summary σ_k appended to a running memory $S_k = S_{k-1} \cup \{\sigma_k\}$, on which the user simulator π_{user} conditions the next round:

$$r_{k+1} \sim \pi_{\text{user}}(\cdot \mid \mathcal{W}_{k+1}, S_k, \mathcal{P}). \quad (1)$$

The round stream is then partitioned into sessions $\mathcal{D} = (D_1, \dots, D_L)$. A two-stage session quality control then repairs persona drift and contradictions: per-chunk audit scans local windows of rounds, while cross-chunk audit catches errors spanning chunk boundaries, with a final expert inspection.

Question Bank Synthesis. From the finalized dataset we synthesize a typed question bank by sampling context windows and prompting an LLM to emit items, each a tuple $q = (x, A, \mathcal{F}_q, \tau)$ with question x , an ordered answer list $A = (a_0, \dots, a_m)$ whose head a_0 is the gold and whose tail gives progressively coarser alternatives for graded scoring, supporting-fact rounds $\mathcal{F}_q$, and type τ . The eight types span three capabilities: *memorizing*—single-session retrieval (ss), file management (FM), and factual judgment (FJ); *reasoning*—multi-session analysis (MS), multimodal reasoning (MR), and temporal reasoning (TR); and *interpreting*—implicit inference (II) and thematic reasoning (TH). Quality validation pass any item that is malformed or cites evidence outside the shown context, so every retained question is answerable from the evidence it cites. Per-type definitions are in Appendix A.3.

2.3 Dataset Statistics

M³Exam comprises 239 multi-session conversations spanning 15 persona scenarios, totaling 3,025 user–assistant rounds paired with 5,150 evaluation questions. The conversations are long-horizon with each persona spanning 14 to 18 sessions with a mean of 12.7 rounds per session, and the conversations together accumulate 1,799 multimodal artifacts.

The question bank is reasoning-heavy and balanced across personas, deliberately weighted toward cases that stress multimodal memory: 1,796 questions (34.9%) are cross-modal (FM, MR), and 769 (14.9%) demand interpretation (TH, II) from unstated context. Breakdown details are shown in Appendix A.1.

3 M³Proctor: Proxy Baseline

Prior multimodal memory methods struggle with the complex, multi-session nature of M³Exam because they are inherently modality-agnostic. Indiscriminately injecting raw visual sources buries decisive evidence and unnecessarily inflates token budgets for queries resolvable by text alone. To facilitate future research, we introduce M³Proctor (Figure 3), a baseline that elevates modality bias (*i.e.*, the specific modality on which a query depends) to a first-class signal. During index construction (§3.1), it projects raw modalities into searchable textual surrogates equipped with modality tags. For modality-aware inference (§3.2), it dynamically detects query bias and employs a cascaded strategy, escalating to raw visual sources only on demand.

3.1 Multimodal Index Construction

To keep the memory searchable by a text retriever while preserving the evidence locked inside images and documents, we project each raw modality into a textual *surrogate* attached to the round that introduced it: images carry a caption, informative visuals such as charts and tables additionally carry a number-rich transcription of their axes, series, and values, and each document page pairs its text layer with a one-sentence digest. Each chunk also carries a modality tag $\#_m(c) \in \{0, 1\}$ over $\mathcal{M} = \{I, P, C\}$ (image, document, chart), read from the surrogate rather than the pixels, which is the zero-cost hook the downstream modules act on. We further attach a compact session-summary chunk to each session which acts as a miniature timeline that preserves

the cross-session signal otherwise fragmented by round-level chunking. Together, the text surrogates and modality tags establish the foundation for the answering cascade (Section 3.2). Because a chart’s surrogate already encodes its underlying quantities, many figure-reading questions are more answerable from textual content alone, allowing the cascade to forgo re-rendering the raw source unless the modality tag and bias signal indicate otherwise.

3.2 Modality-Aware Inference

At inference time, M³Proctor exploits the indexed modality tags and the question’s *modality bias* to determine, for each retrieved chunk, whether its low-cost textual surrogate suffices or the raw source must be consulted. This process carries out in three coupled steps: M³Proctor first detects the question’s modality bias and applies it to re-rank the retrieved evidence, surfacing the most relevant dialogues irrespective of modality. It then attempts an answer from the textual surrogates of the dialogues alone, resorting to the raw visual or document sources only when this text-surrogate attempt is deemed unreliable.

Bias detection and bias-aware re-ranking. We make modality demand explicit by representing the bias of a question as a binary vector indexed by $\mathcal{M}$,

$$b(q) = (b_m(q))_{m \in \mathcal{M}} \in \{0, 1\}^{|\mathcal{M}|}, \quad (2)$$

where $b_m(q) = 1$ iff answering q requires inspecting a source of modality m . A single instruction-tuned LLM call returns $b(q)$, and the result is cached so this overhead is paid at most once per question—far cheaper than a recurring multimodal forward pass. Crucially, $b(q)$ is a *shared* signal that both re-ranking and answering consume, keeping modality decisions consistent end to end. Re-ranking is its first use. Dense retrieval ranks chunks by semantic similarity alone, which can bury a round that holds the decisive image but is only moderately similar to q ; we therefore use $b(q)$ to correct the ranking *within* the candidate pool. We first retrieve a pool of the chunks with highest cosine similarity $\cos(q, c)$, then re-rank each candidate by a bias-modulated score as follows

$$s(q, c) = \cos(q, c) + \sum_{m \in \mathcal{M}} \lambda_m b_m(q) \mathbb{1}_m(c), \quad (3)$$

and keep the k highest-scoring chunks. The bonus λ_m applies only when the question is biased toward m ($b_m(q) = 1$) while the chunk carries that

modality ($\mathbb{1}_m(c) = 1$); we set $\lambda_p > \lambda_I > \lambda_C$ since document-bearing chunks are rarest and most diagnostic, and add a lightweight diversity constraint so the retained set covers the session-summary view and spans more than one session, protecting multi-session and temporal questions (details in Appendix F). Because the bonuses are small relative to the cosine range, re-ranking breaks ties in favor of the right modality but cannot promote an unrelated chunk, and it operates entirely on index-time tags and surrogates, so improving the modality composition of the top- k incurs *no* multimodal cost with raw sources staying untouched at this stage.

Cost-aware modality cascading. Re-ranking decides which evidence to read, then the cascade decides how much to pay for it. Since the decisive cost in a multimodal assistant is incurred exactly when raw sources enter the prompt, yet a chunk’s textual surrogate often already suffices, we make answering a cascade that prefers the cheapest sufficient representation and pays for raw modalities only on demand. Stage 1 answers q from text only, using generated surrogates including captions, transcriptions, and file digests presented in the retrieved context. A confidence test Φ then decides whether this answer can be trusted, escalating ($\Phi=1$) when the text answer is unreliable ($u(\hat{a}_1)=1$) or when the modality bias is unsatisfied by text ($\beta(q)=1$):

$$\Phi(\hat{a}_1, q) = \mathbb{1}[u(\hat{a}_1) \vee \beta(q)]. \quad (4)$$

If reaches escalation threshold, Stage 2 attaches the raw source of modality m under a per-modality gate driven by a fused modality-evidence score

$$e_m(q) = w_r \frac{1}{k} \sum_{c \in \mathcal{R}_k} \mathbb{1}_m(c) + w_b b_m(q) + w_s \sigma_m(q), \quad (5)$$

which fuses, respectively, the fraction of retrieved chunks carrying modality m , the detected bias, and a surface lexical cue $\sigma_m(q)$; the raw source is supplied only when this evidence clears the threshold.

The common case of a text-answerable question terminates after one cheap call, while the modality gate ensures that raw modality sources are consumed only when fused evidence, rather than a single brittle cue, indicates they are necessary. This yields the trade-off we target: questions that genuinely require inspection still escalate and retain accuracy, whereas text-answerable questions avoid the multimodal premium, lowering the average token cost without sacrificing correctness.

Table 2: Overall performance comparison on M³Exam. All memory systems are implemented with Qwen-2.5-VL-7B as the default backbone. Best and second-best are marked in **bold** and underline: Frontier MLLMs are ranked separately, while text-based and multimodal agentic-memory systems are ranked as a single group.

Agent Type	SS		MS		TR		MR		TH		II		FM	FJ	Overall Score
	F1	LLM	F1	LLM	F1	LLM	F1	LLM	F1	LLM	F1	LLM	EM	EM	
Frontier MLLMs															
🌟 Claude-Opus-4.6	0.2593	0.6050	0.3678	0.5687	0.3754	<u>0.5813</u>	0.5354	0.7306	0.2387	<u>0.6083</u>	0.2412	0.6501	<u>0.2203</u>	0.9443	0.4937
🌀 GPT-5.4	0.5257	<u>0.7623</u>	0.3276	0.6274	0.4975	0.4950	0.4477	0.6389	0.1683	0.4333	0.3227	0.7125	0.1890	0.9621	0.4823
🔢 GLM-5.1	<u>0.5409</u>	0.6510	0.6361	0.6066	0.5142	0.5115	0.5515	0.6550	0.5843	0.6193	0.3735	0.5667	0.1667	0.9332	0.5493
👤 Qwen3.6-Plus	0.4713	0.6354	0.5907	0.6004	<u>0.5218</u>	0.4799	0.5167	0.6406	0.5325	0.5227	<u>0.3347</u>	0.3833	0.1520	0.9800	<u>0.5049</u>
🌟 Gemini-3.1-Pro-Preview	0.5320	0.7693	0.2991	0.5437	0.4926	0.5404	0.3789	0.5111	0.3161	0.4417	0.2870	<u>0.6875</u>	0.2403	<u>0.9666</u>	0.4668
🌟 Doubao-Seed-2.0-Pro	0.5638	0.2917	<u>0.6330</u>	0.3607	0.6514	0.6437	0.7204	0.6586	<u>0.5480</u>	0.1364	0.3005	0.4103	0.1950	0.9532	0.4938
👤 Kimi-k2.5	0.4269	0.6304	0.3410	0.4536	0.4642	0.5050	<u>0.5614</u>	0.6556	0.1987	0.4351	0.2487	0.5589	0.1308	0.8667	0.4493
🌀 Grok-4	0.4383	0.6211	0.3129	<u>0.6125</u>	0.3259	0.5150	0.3912	<u>0.7034</u>	0.1450	0.3667	0.2322	0.6875	0.1871	0.7923	0.4357
👤 Qwen-2.5-VL-7B	0.2742	0.5468	0.4587	0.3471	0.4378	0.5063	0.5425	0.4682	0.2461	0.0635	0.1672	0.0523	0.2070	0.7904	0.4179
Text-based Agentic-Memory Systems															
NaiveRAG	0.4430	0.6547	0.3117	0.4429	0.2349	0.4167	0.2080	0.2572	0.1321	0.2806	0.1263	0.3202	0.2569	0.7840	0.3249
A-Mem	0.5921	<u>0.7247</u>	0.3998	0.5345	0.1298	0.1697	0.2403	0.3727	0.1361	0.3493	0.1227	0.2872	0.2875	0.9243	0.3450
Mem0 (w/o vision)	0.5343	0.6957	0.3837	0.5448	0.1361	0.1853	0.2471	0.3787	0.1346	0.3843	0.0962	0.3000	0.1544	0.9243	0.3410
MemoryOS	0.5410	0.7047	0.3399	0.5145	0.1248	0.3977	0.2146	0.3299	0.1274	0.3327	0.1162	0.3175	0.3150	<u>0.9733</u>	0.3470
Multimodal Agentic-Memory Systems															
Universal-RAG	<u>0.5858</u>	0.7328	0.4349	0.5769	0.3903	0.4431	0.2938	0.4369	0.2691	<u>0.4872</u>	0.1586	0.5559	0.5367	0.8241	0.4366
RAG-Anything	0.5849	0.6418	0.3940	0.5474	<u>0.4353</u>	0.4465	0.3863	0.4970	<u>0.2782</u>	0.4390	<u>0.1873</u>	<u>0.6201</u>	0.4572	0.8508	0.4481
Mem0 (with vision)	0.4798	0.6457	<u>0.4181</u>	0.5548	0.1511	0.1837	0.4459	0.5607	0.1485	0.3527	0.1286	0.5973	0.1942	0.9176	0.3969
MemVerse	0.4850	0.6406	0.3895	0.5662	0.3229	0.4096	<u>0.4591</u>	0.5613	0.2518	0.4223	0.1849	0.6048	0.5474	0.9042	0.4481
NGM	0.3559	0.5266	0.3453	0.4660	0.1716	0.1601	0.4708	0.5479	0.2288	0.3905	0.1807	0.6060	0.6208	0.9042	0.3813
MIRIX	0.4954	0.6378	0.2853	0.4875	0.4564	0.4949	0.4076	<u>0.5775</u>	0.2999	0.5917	0.1054	0.5375	0.3807	0.8664	<u>0.4560</u>
M ³ Proctor (Ours)	0.4993	0.6924	0.3697	<u>0.5667</u>	0.3952	<u>0.4877</u>	0.4305	0.6063	0.2202	0.4786	0.3021	0.6515	<u>0.5688</u>	0.9844	0.4838

4 Benchmarking Analysis

4.1 Experimental Setup

Models and Baselines. We compare two families of systems on M³Exam. *Frontier MLLMs* include the latest strong closed-source models—Claude-Opus-4.6, GPT-5.4, GLM-5.1, Qwen3.6-Plus, Gemini-3.1-Pro-Preview, Doubao-Seed-2.0-Pro, Kimi-k2.5 and Grok-4—answering directly from the labeled supporting evidence without external memory module, serving as a no-memory reference (details in Appendix B.1). *Agent-memory systems* instead manage an external memory and retrieve from it; we group them into text-based methods: NaiveRAG (Lewis et al., 2020), A-Mem (Xu et al., 2025), Mem0 (Chhikara et al., 2025), and MemoryOS (Li et al., 2025), and multimodal methods: UniversalRAG (Yeo et al., 2025), RAG-Anything (Guo et al., 2025), MemVerse (Liu et al., 2025), NGM (Fisher, 2025), MIRIX (Wang and Chen, 2025), and Mem0 with vision enabled. All agentic-memory systems, including M³Proctor, are deployed on the same Qwen-2.5-VL-7B (Bai et al., 2025) backbone (base results highlighted). Implementation details are deferred to Appendix F.

Evaluation Metrics. We evaluate different model answers under four complementary metrics: exact match (EM), token-level F1 score, BLEU-1, and the LLM-as-a-Judge (LLM-J) adapted from Mem-Gallery (Bei et al., 2026) (using Qwen2.5-VL-32B-Instruct as the judge) to report representative metrics. Since FM (image identifiers) and FJ (single-letter choices) have closed-form answers, we only report the EM metric. We further summarize each model with an *Overall Score*, a weighted comprehensive metric over all evaluated answer-quality signals. Complete scoring definitions and implementation details can be found in Appendix B.2.

4.2 Main Results

Table 2 reports the main results with result breakdown and alternative backbone evaluations in Appendix C.1. We read it along three questions: how far current systems are from solving M³Exam, where they fail, and whether M³Proctor closes the gap.

❶ *M³Exam poses a significant challenge.* The strongest frontier MLLM (GLM-5.1) reaches only 0.549 overall, *even when relieved of retrieval* (full-context variant shown in Appendix C.2 degrades further still). Agentic-memory systems (built on

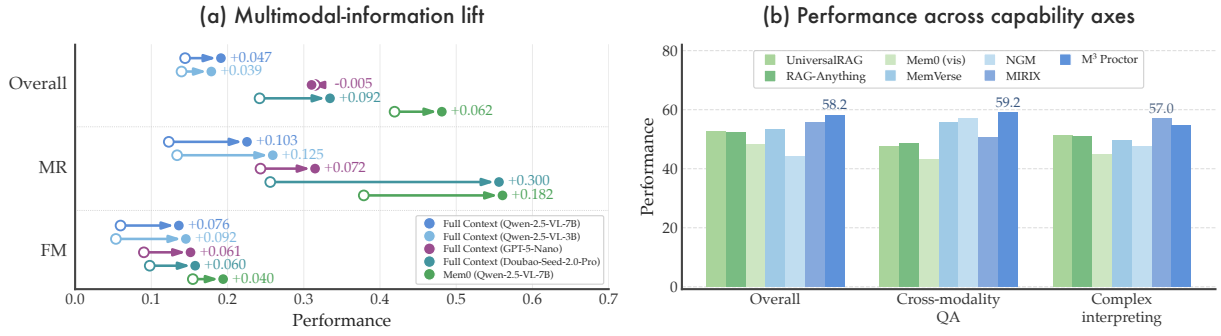


Figure 4: Multimodal influence and capability profile on M³Exam. *Cross-modality QA* involve MR, FM and other queries labeled as modality-coupled, while *Complex Interpreting* involve TH and II.

Qwen-2.5-VL-7B) improve over the base model but generally trail frontier MLLMs, with multimodal-enhanced variants scoring higher. Taken together, these results indicate that M³Exam remains far from solved, exposing capabilities that current models and memory systems have yet to acquire.

② *Cross-modal and implicit-intent questions are the bottleneck.* Results show that systems handle single-session reasoning well but degrade sharply on the capabilities M³Exam targets. The base model collapses on complex interpretation tasks (0.064 on TH and 0.052 on II) and remains weak on cross-modal reasoning (0.468 on MR and 0.207 on FM), yet external memory recovers much of this shortfall, approaching and at times surpassing frontier MLLMs. These results suggest that the bottleneck of M³Exam lies in realistic multimodal reasoning and implicit interpretation, which stems from inadequate evidence management rather than a fundamental backbone ceiling and is therefore addressable.

③ *M³Proctor advances the agentic-memory frontier.* M³Proctor attains the best Overall Score among agentic-memory systems (0.484 vs. 0.456 for the prior best), with the gains concentrated where the benchmark is hardest: it leads on MR (0.606 LLM-J), II (0.652), and FM (0.569 EM). Most notably, despite running on a modest 7B backbone and retrieving its own evidence, M³Proctor surpasses several retrieval-relieved frontier MLLMs and trails only the strongest proprietary models. This positions M³Proctor as a competitive agentic-memory system that matches frontier MLLMs in multimodal memory while operating at a fraction of their scale.

Takeaway : M³Exam remains far from solved, with cross-modal grounding and implicit inference as the core challenge, but M³Proctor offers an efficient open-source baseline.

5 In-Depth Analysis

We now further analyze the multimodal dependence of M³Exam (Figure 4), and the cascade behavior and efficiency performance of M³Proctor (Figure 5).

① *M³Exam is tightly coupled to its visual evidence.* Figure 4 (a) measures the multimodal-information lift: the performance gain a system obtains from accessing raw multimodal evidence rather than its generated textual surrogates alone. The lift is modest overall but pronounced on the strongly multimodal subsets, reaching up to +0.300 on MR and +0.116 on FM, indicating that a textual surrogate cannot stand in for the visual source on these questions. The capability profile in Figure 4(b) is consistent: systems differ most on cross-modality QA, where M³Proctor leads at 0.592 and remain separated on complex interpreting (0.570), confirming that the benchmark’s hardest questions genuinely hinge on multimodal evidence and implicit inference rather than lexical matching over the dialogue text.

② *The modality cascade fires where, and only where, vision is needed.* Figure 5 (a) shows that the cascade triggers far more often on visual types (68.0% on FM, 51.1% on MR) than on text types ($\leq 15\%$), indicating that modality-bias detection reliably identifies which queries require a raw source. Figure 5 (b) confirms the escalation pays off precisely there: activating the cascade lifts answer quality by 0.116 on FM and 0.092 on MR while leaving text types essentially unchanged. The cascade thus supplies multimodal evidence selectively, rather than re-feeding it indiscriminately.

③ *M³Proctor matches multimodal accuracy at text-only cost.* Figure 5 (c) plots index-build time against per-query token consumption, where cascade escalation calls are taken into account. Results show that although M³Proctor is a multimodal memory, its footprint falls within the *text-only* regime—72 s to

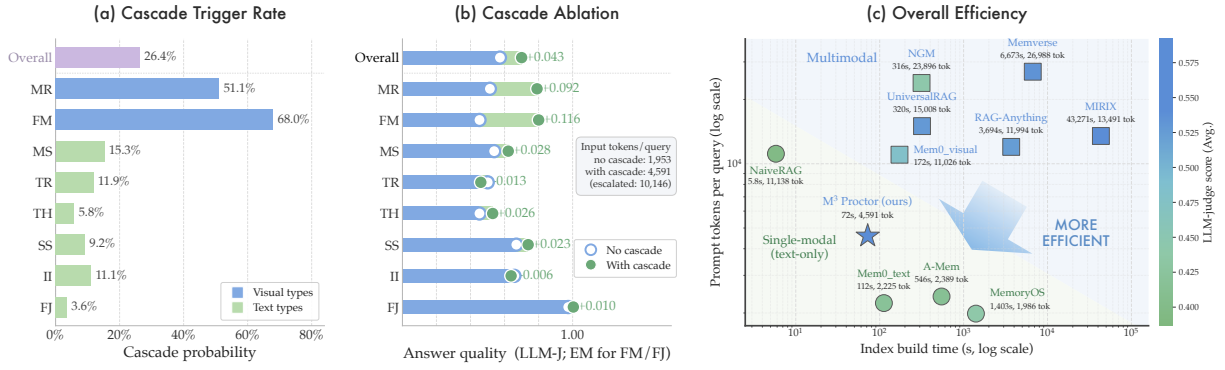


Figure 5: In-depth analysis of cascade ablation and M³Proctor efficiency performance.

build the index and 4,591 tokens per query—one to three orders of magnitude below multimodal baselines such as MIRIX (43,271 s) and NGM (26,988 tokens), since raw visuals enter the prompt only on the 26.4% of queries that the cascade escalates. M³Proctor nonetheless attains the highest performance of any system, demonstrating that modality-aware escalation—rather than indiscriminate visual injection—is what delivers multimodal accuracy.

Takeaway : M³Exam relies heavily on its visual evidence, and M³Proctor’s modality cascade supplies that evidence only when needed—matching multimodal level accuracy at significantly reduced cost.

6 Related Work

Benchmarks for Long-term Multimodal Memory. Early efforts ground multi-turn QA in documents, with MMLongBench-Doc (Ma et al., 2024) over rendered pages and MultiDoc2Dial (Feng et al., 2021) yet treat documents as static inputs rather than accumulating memory. Later benchmarks shift to evolving dialogue histories: LongMemEval (Wu et al., 2025) targets retrieval across very long histories, while MemoryArena (He et al., 2026) and MemoryAgentBench (Hu et al., 2025) probe fact update, consolidation, and retrieval, but remain text-only. LoCoMo (Maharana et al., 2024) introduced images into long-term conversational memory, and recent multimodal benchmarks like MM-Dialog (Feng et al., 2023), MMRC (Xue et al., 2025), Mem-Gallery (Bei et al., 2026) push toward multi-session multimodal memory, yet treat images only as topical anchors and leave out implicit intent. M³Exam unifies these strands, uniquely evaluating implicit-intent inference over realistic accumulating document-level multimodal conversations.

Systems for Long-term Multimodal Memory.

To overcome fixed context windows, external memory systems store past interactions and retrieve relevant evidence on demand. Textual memory systems like A-Mem (Xu et al., 2025) links interactions into a Zettelkasten-style note network, while Mem0 (Chhikara et al., 2025) consolidates salient facts into a compact store, and MemoryOS (Li et al., 2025) unifies parametric, activation, and plaintext memory under an OS abstraction. The inspired multimodal memory systems like UniversalRAG (Yeo et al., 2025) and RAG-Anything (Guo et al., 2025) which retrieve across text, images, and tables; MemVerse (Liu et al., 2025) and neural graph memory (Fisher, 2025) encode cross-modal associations; and MIRIX (Wang and Chen, 2025) manages multimodal experience through specialized components. However current systems conduct global uniform retrieval, regardless of whether a query actually needs visual evidence grounding. M³Proctor predicts a query’s multimodal intent and returns text surrogates or raw artifacts accordingly, recovering evidence at minimal cost.

7 Conclusion

We introduce M³Exam, a query-centric benchmark evaluating how well multimodal conversational agents memorize, reason, and interpret over long-horizon histories of dialogue, images, and documents. Our evaluation reveals a clear gap: existing memory systems and frontier MLLMs handle single-session recall but struggle with cross-modal grounding, cross-session reasoning, and implicit-intent inference. We therefore propose M³Proctor, a modality-aware method that infers a query’s modality bias and escalates to raw sources only on demand, improving accuracy while substantially cutting tokens and index-construction time.

Limitations

Although M³Exam targets realistic multimodal user-agent interactions, the current benchmark mainly focuses on single-turn QA over accumulated history. Real-world interactions often involve long-horizon multi-turn conversations with evolving intents and iterative memory updates, which remain underexplored in all existing multimodal memory benchmarks. We plan to further extend M³Exam toward more dynamic multi-turn memory evaluation in future work. In addition, while M³Proctor improves modality-aware multimodal retrieval and reasoning efficiency, its capability on interpreting-oriented questions remains limited. Better modeling implicit user states and performing deeper contextual interpretation over long-term multimodal interactions remain important directions for future work.

Ethics Statement

Data provenance and privacy. M³Exam contains no real user data: every persona, core event, timeline, and dialogue turn is synthetically generated by large language models from hand-authored seeds (Table 8), so the conversations describe fictional individuals. The personas are occupationally varied yet avoid sensitive attributes and real-world identities; any resemblance to an actual person is unintended. Being fully synthetic, the benchmark raises no consent, de-identification, or PII concerns.

Multimodal artifacts. The visual and document evidence attached to each conversation is an assembled persona-aligned file pool of charts, photographs, and PDF documents. We use only artifacts that are obtained under terms that permit research use, and we do not redistribute copyrighted third-party material. We manually screened the generated content to remove any unsafe, offensive, or privacy-leaking material before release.

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APPENDIX

M³EXAM: BENCHMARKING MULTIMODAL MEMORY FOR
REALISTIC USER-AGENT INTERACTIONS

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A Dataset Details and Statistics

A.1 Dataset Statistics

Figure 6 summarizes the per-persona scale of the benchmark. In aggregate, M³Exam comprises 239 multi-session conversations spanning 3,025 user–assistant rounds, 1,799 attached images and multimodal artifacts, and 5,150 evaluation items, of which 1,975 (38.3%) are cross-modality questions whose answers require grounding in an image or file. The corpus is deliberately long-horizon: each persona holds 14–18 sessions and 163–256 dialogue turns (Panel A), so answering a question typically demands recalling content deposited across many earlier sessions rather than within a single recent context window. The footprint is also balanced across personas (Panel B): every persona contributes a comparable number of QA items (320–362) and attached files (107–133), and the nine document-bearing personas (bottom block) carry a markedly higher share of cross-modality questions

(≥ 141 each) than the six dialogue-only personas, reflecting the extra chart- and PDF-grounded items their later sessions support. This even, large-scale coverage ensures that no capability is measured on only a few scenarios.

A.2 Data Distribution

Figure 7 characterizes M³Exam along four axes that motivate its use as a cross-modal memory benchmark.

Answers are non-local and reach far back in the history. Panel (1) quantifies how far apart in the conversation a question’s supporting evidence lies. Of the 1,510 cross-session questions, 72% require evidence from rounds at least two sessions apart, 42% span five or more sessions, and the distribution carries a long tail out to a maximum of 17 sessions. Because each persona holds 14–18 sessions, a model cannot answer these items from recent context alone: it must retain and retrieve information deposited many sessions earlier. This long-range structure is the property that text-summarisation or recency-biased baselines fail to capture, and it complements the high *Cross-Modality QA* counts in Figure 7.

Dialogues exhibit a realistic, asymmetric exchange. Panel (2) shows that user and assistant turns occupy clearly different length regimes. User turns are short and tightly concentrated (median 29 words, mean 37), reflecting focused questions and follow-ups, whereas assistant turns are substantially longer and heavier-tailed (median 57, mean 85, up to 379 words), reflecting detailed explanations and multi-step procedures. This give-and-take mirrors genuine assistant usage rather than balanced chit-chat, and the length of assistant turns is what makes single-session recall (ss) and thematic (TH) questions non-trivial.

Coverage is systematic across all scenarios. Panel (3) reports the per-persona breakdown of the eight question types. The grid is fully populated—every one of the 15 personas contributes every type—so no capability is evaluated on only a handful of scenarios. The heatmap also makes the benchmark’s reasoning emphasis visible: multi-modal reasoning (MR) is the largest column for the document-bearing personas (up to 100–102 items each), consistent with the construction pipeline allocating extra chart- and PDF-grounded questions to personas whose later sessions include documents.

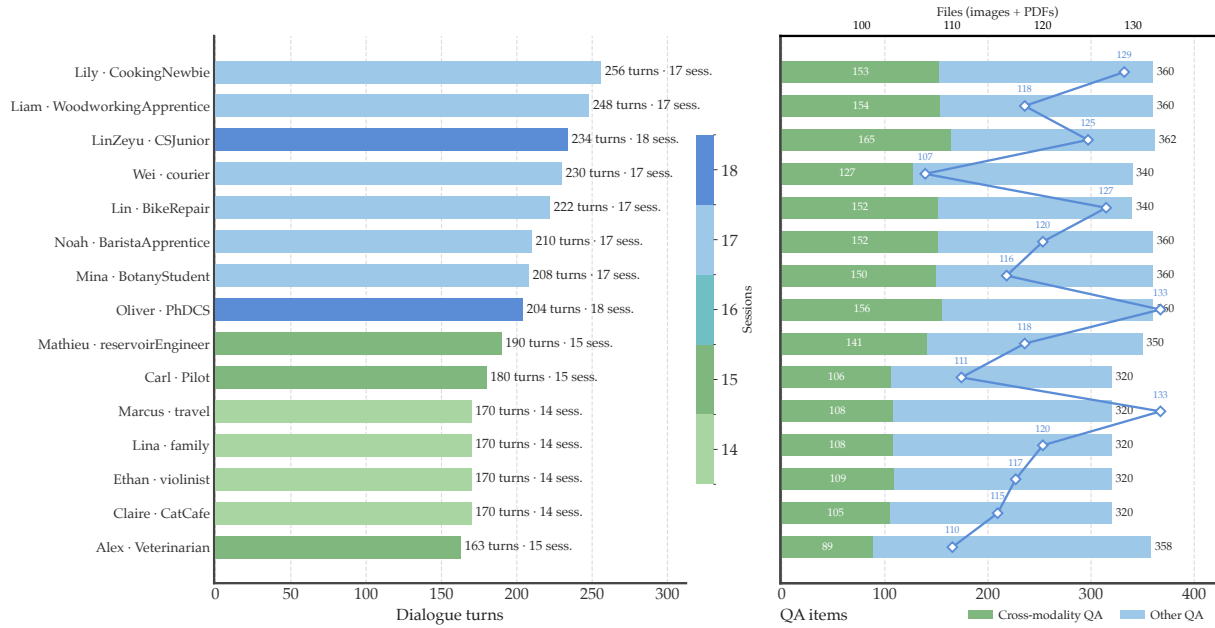


Figure 6: **Per-persona scale of M³Exam.** (A) *Conversation size.* Dialogue turns per persona, with bar colour encoding the number of sessions. (B) *QA and asset footprint.* Total QA items per persona, split into cross-modality QA (green; FM and MR) and the remaining QA (blue); the overlaid line (top axis) reports attached files per persona.

Sessions are spread irregularly over real time. Panel (4) plots each persona’s sessions on a normalized calendar axis. Rather than being back-to-back, the 239 sessions are distributed unevenly—with bursts and multi-week gaps—over spans ranging from 8 to 127 days (a mean of 12.7 turns per session). This temporal spread is precisely what gives the timeline-grounded temporal-reasoning questions (TR) their difficulty, since “when did X happen” cannot be answered by turn order alone.

A.3 Question Types and Synthesis

Figure 8 summarizes the composition and difficulty of the 5,150 evaluation items. Panel (A) shows that the benchmark is reasoning-heavy: multimodal reasoning (MR, 1,142 items, 22.2%), temporal reasoning (TR, 787, 15.3%), and single-session recall (ss, 748, 14.5%) are the three largest types, followed by figure management (FM, 654, 12.7%) and multi-session (ms, 601, 11.7%); the interpreting types—thematic reasoning (TH, 464, 9.0%), factual judgment (FJ, 449, 8.7%), and implicit inference (II, 305, 5.9%)—are deliberately rarer but the hardest to construct, since they hinge on unstated context. No single type dominates, so accuracy on the benchmark cannot be inflated by mastering one easy category. Panel (B) places each type by its average evidence load (x : number of supporting facts) and memory span (y : number of distinct sessions the evidence spans), with bubble radius

growing with the fraction of items needing visual or PDF grounding. The types separate along both axes: ss sits in the low-load, single-session corner, whereas II and ms demand the widest memory span and MR and FM carry the strongest visual grounding (largest bubbles). This spread confirms that M³Exam jointly stresses long-range memory and cross-modal grounding rather than any single dimension.

M³Exam generates eight typed question categories on each finalized dataset. Each item carries a question, an ordered list of answers (the first entry is the gold; the remaining entries are progressively coarser alternatives used by the graded EM metric; see Section 4.1), and round-level supporting_facts that identify the dialogue rounds an evaluator must surface for the model. All types share the same outer loop: we sample a dialogue context (one or more session windows), fill a type-specific prompt template, call an LLM to emit a JSON array of items, validate each item against the schema in Section 2.2, and append the surviving items to question.json; generation resumes from any partial bank until a per-type target count is met. Below we describe each type along two axes—*what it tests* and *how it is synthesized*, the latter giving the sampling strategy and the construction-time constraint that defines the type. The illustrative items are drawn verbatim from the Alex_Veterinarian dataset (*persona*: a senior veterinary clinician); the corresponding

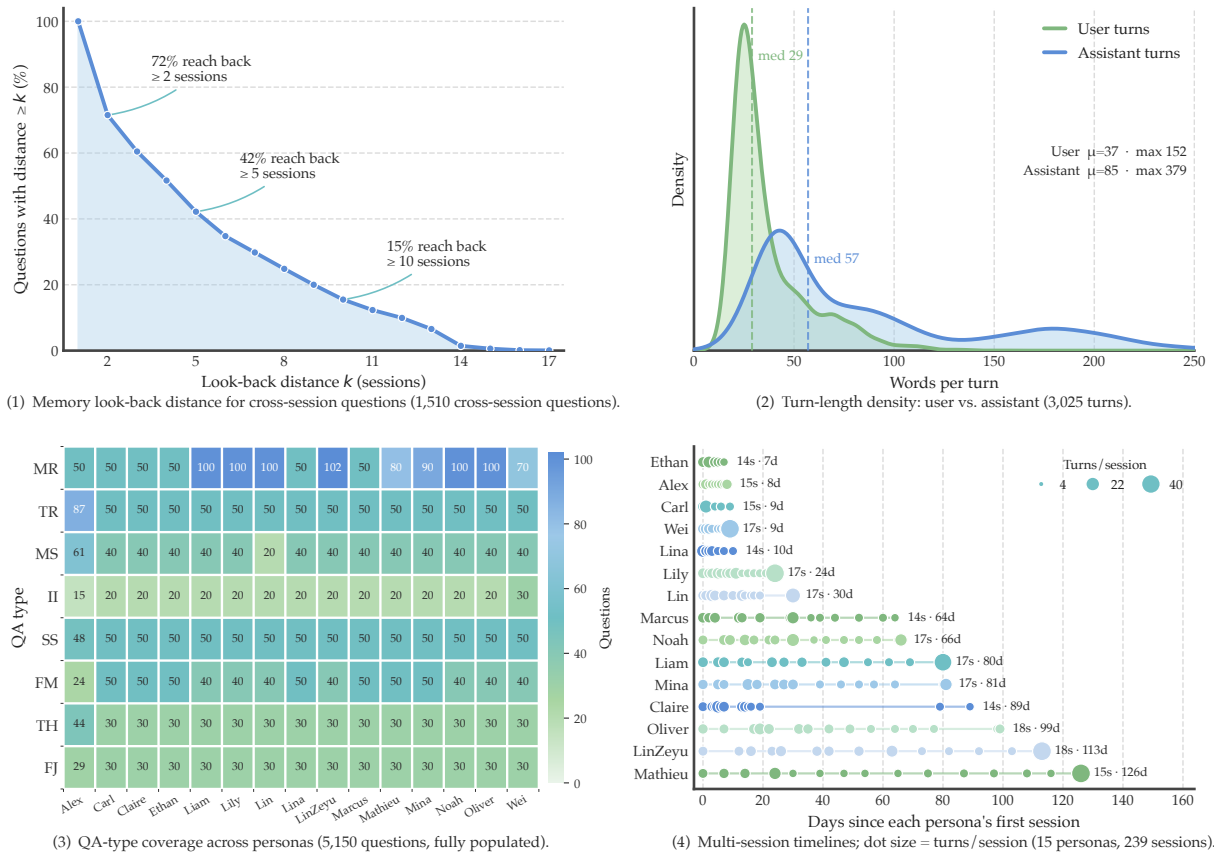


Figure 7: **Distributional properties of M³Exam.** (1) *Memory look-back distance.* For each cross-session question, the session-index gap between the earliest and latest round it cites in `supporting_facts`; the curve plots the fraction of questions whose evidence spans at least k sessions. (2) *Turn-length density.* Gaussian-kernel density of word counts for user versus assistant turns. (3) *QA-type coverage across personas.* Number of questions of each type (rows) per persona (columns). (4) *Multi-session timelines.* One track per persona, each dot a session at its real calendar date (normalized to day 0), with dot size encoding turns per session.

verbatim prompt templates follow in Section E.

SS — Single-Session. SS questions test detailed recall from a single dialogue session: the gold answer is grounded in one round, and a model that has not read that session must not be able to answer correctly. *Synthesis.* We sample a contiguous window of 4 to N rounds from a single session and ask the generator to write questions answerable *only* from that excerpt. The construction constraint is locality: the supporting fact must cite exactly one round, and the gold answer is kept to a short phrase (≤ 3 words) so the item is unambiguous and recall-focused.

MS — Multi-Session. MS questions require connecting or comparing information across *at least two* sessions; a question answerable from a single session alone is rejected at generation time. *Synthesis.* We sample 2 to 3 distinct sessions and draw a window from each. The generator must write a

question whose answer requires fusing facts from at least two different sessions, so the supporting facts cite ≥ 2 rounds drawn from different sessions (e.g., D2:5, D16:9).

TR — Temporal Reasoning. TR questions hinge on dates, ordering, or time spans drawn from the event timeline and session headers. *Synthesis.* We sample 2 to 4 sessions and sort them chronologically; questions hinge on absolute dates, event ordering, or time-range counts taken from the timeline and session headers. To absorb formatting variation, the gold answer is stored in multiple date formats (e.g., ["2023-10-10", "October 10, 2023", "Oct 10 2023"]) so that partially correct phrasings receive partial credit under the graded EM, and the supporting facts include the recap round bearing the absolute date.

MR — Multimodal Reasoning (chart-grounded). MR questions require reading a number, label, rank,

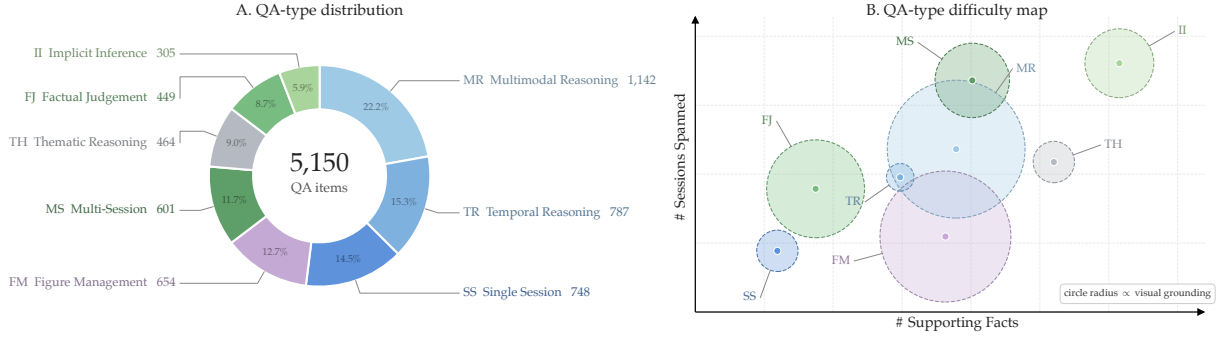


Figure 8: **QA-type composition and difficulty of M³Exam.** (A) *Distribution.* Number and share of the 5,150 QA items across the eight question types. (B) *Difficulty map.* Each type is placed by its average evidence load (x: number of supporting facts) and memory span (y: number of distinct sessions the supporting facts span), with bubble radius growing with the fraction of items requiring visual or PDF grounding. Colors denote the question type and are shared across both panels.

or simple arithmetic relation from a visible chart. *Synthesis.* We sample a single session centred on image-bearing rounds; for PDF sessions, the relevant pages are rendered and attached so the vision model sees them directly. The defining constraint is that the answer is visible in the attached image or page but *not* present in the dialogue text, which forces genuine visual grounding rather than text-only recall. The supporting facts cite the image-bearing round.

FM — Find-Matching Image. FM questions ask the model to identify which image in the dataset matches a textual description; the answer is the image file name (in `img_<number>` format), testing cross-modal retrieval over the dataset’s flat image folder. *Synthesis.* We sample a round that exposes at least two candidate images and ask the generator to write a description that uniquely matches exactly one of them. The answer list places the matching image identifier first, followed by the competing candidates (e.g., [`"img_60"`, `"img_61"`]), so the task is cross-modal retrieval under hard negatives.

TH — Thematic Reasoning. TH questions test recall of multi-step procedures, decision rationales, or case-management protocols that the assistant introduced during the dialogue; they typically have a structured gold answer (e.g., a 5-pillar treatment plan) and reward partial recall via the answer-list gradient. *Synthesis.* We sample a single session with a wider window (5 to 8 rounds) and ask the generator to bundle at least three decisions or steps from one case thread into one structured answer, stored hierarchically (full $\rightarrow$ short $\rightarrow$ keyword) so the graded metric rewards partial recall of the procedure.

II — Implicit Inference. II questions require inferring a fact that is never stated directly, by composing implicit signals from multiple rounds (e.g., role, relationship, or motivation); supporting facts cite the rounds that together justify the inference. *Synthesis.* We sample up to five sessions with short windows each. The generator must write a question whose answer is *never stated outright* but can be inferred by composing ≥ 2 clues spread across the history; the answer is a qualitative noun phrase, and frequency-count answers are disallowed to keep the item inference-driven rather than tally-driven. Detailed case study shown in Appendix C.3.

FJ — Factual Judgment. FJ questions are multiple-choice with one correct option and three plausible distractors, covering yes/no judgments and choose-the-better-recommendation decisions, and are scored under EM with a single-letter gold answer. *Synthesis.* We sample a multi-session excerpt and embed four options (A–D) directly in the question text, with distractors drawn from the same corpus to make them plausible. The generator is instructed to output only the gold letter, yielding an unambiguous multiple-choice item.

A.4 Quality Control.

To prevent errors and hallucinations from propagating downstream, we apply quality control at every stage, combining an automatic LLM self-check with expert verification. For timeline generation, the self-check audits persona violations, core-event drift, offensive content and internal contradictions (e.g., date-order or mutually exclusive states). For conversation synthesis, we audit at two granularities: a per-chunk audit scans windows of rounds for

inconsistencies, while a cross-chunk audit catches those spanning chunk boundaries, and each detected flawed round is rewritten by a repair pass conditioned on neighboring rounds. For the question bank, a validation pass enforces well-formedness and evidence consistency, discarding any question whose supporting facts fall outside the sampled context window so that every retained question is answerable from the evidence it cites. These edits made across these stages are documented, and a final expert inspection of the records verifies persona fidelity and core-event coverage. Prompts and the implement details are given in Appendix A.3.

A.5 Persona and Core-Event Specifications

Each persona scenario in M³Exam is seeded from a *persona* (who the user is and what their interactions cover) and a *core event* (the storyline that the multi-session timeline elaborates). Table 8 summarizes both seeds for all fifteen subsets; these are the inputs to the timeline-construction prompts described in Section E.

B Experimental Details

B.1 MLLM Details

Implementation The frontier MLLMs in the main results (Table 2) have no external memory module. They answer directly from the evidence placed in their context. For a fair comparison that tests reasoning rather than retrieval, we do not dump the entire conversation history into the window. Instead, for each question we follow its annotated supporting facts to the sessions the answer depends on and supply those sessions together with their attached images and rendered PDF pages, so every model sees the same gold evidence. This implementation removes the retrieval burden and upper-bounds what each model could achieve given perfect memory.

Long-context buffer baseline. Modern language models often support extended context windows ranging from 128K to over 1M tokens, which suggests a simple memory-free strategy: keep a buffer of the most recent tokens. In a model with a 128K-token limit, the agent concatenates incoming chunks until the total exceeds the window size; once the limit is reached, the earliest chunks are evicted in a first-in-first-out (FIFO) manner. Such an agent relies solely on positional recency and assumes the model can attend effectively over whatever currently fits in its window. We include this buffer behavior

as a reference point for the no-memory regime, in contrast to the oracle-context feeding above.

Models. We evaluate the following frontier closed-source MLLMs as answering models. *Claude-Opus-4.6* (Anthropic, 2026a) is Anthropic’s flagship reasoning model. *GPT-5.4* (OpenAI, 2026b) is OpenAI’s latest general-purpose multimodal model. *Gemini-3.1-Pro* (Google DeepMind, 2026) is Google’s multimodal model with a very long native context window. *GLM-5.1* (Zhipu) (Zhipu AI, 2026) and *Qwen3.6-Plus* (Alibaba) (Qwen Team, 2026) are strong open-weight-lineage multimodal models. The arena additionally includes *Doubao-Seed-2.0* (ByteDance) (ByteDance, 2026), *Kimi-k2.5* (Moonshot) (Moonshot AI, 2026), and *Grok-4* (xAI) (xAI, 2025). All API-served models are accessed through OpenAI-compatible remote endpoints and exposed to the same function-calling interface. For the full agentic-memory baseline experiments, we additionally evaluate all memory systems on four answering backbones spanning families and scales. The lightweight/open-weight group includes Qwen2.5-VL-7B and Qwen2.5-VL-3B (Qwen Team et al., 2025), and two proprietary models, GPT-5-nano (OpenAI, 2026b) and Doubao-Seed-2.0-Pro, to confirm that our findings are not an artifact of a single backbone.

B.2 Metrics

We split evaluation into two metric families: *answer-quality* metrics that score the model’s predicted answer against the gold, and *retrieval-quality* metrics that score how well a retrieval-augmented baseline surfaces the supporting-fact rounds annotated in our question bank. Let $\hat{a}$ denote the predicted answer and $\mathcal{A} = (a_0, a_1, \dots)$ the ordered gold answer list, where a_0 is the canonical gold (identical to the label field in the dataset) and each a_i for $i>0$ is a progressively coarser alternative phrasing. Because exact and soft matching call for different text normalization, we use two operators: $\text{norm}_{\text{em}}(\cdot)$ is a *sentence-level* normalization (lowercasing, collapsing runs of whitespace to a single space, and stripping leading/trailing non-alphanumeric-underscore characters) used for whole-string equality, while $\text{tok}(\cdot)$ is the SQuAD-style normalization (lowercasing, punctuation stripping, article removal) followed by whitespace tokenization, used for F1 and BLEU-1.

Graded Exact Match (EM). We follow long-term-memory benchmarks (Maharana et al., 2024; Bei et al., 2026) that use exact-match scoring on short factual answers, and extend it into a gradient over the ordered gold list. Under sentence-level normalization we take the earliest matching position $i^* = \min\{i : \text{norm}_{\text{em}}(\hat{a}) = \text{norm}_{\text{em}}(a_i)\}$ and assign a discounted score:

$$\text{EM}(\hat{a}, \mathcal{A}) = \begin{cases} 2^{-i^*} & \text{if } i^* \text{ exists,} \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

so predicting the canonical gold scores 1.0, predicting the first alternative scores 0.5, and so on. Compared with binary EM, this rewards near-correct but less specific phrasings in proportion to how close they are to the canonical answer, while still preferring the canonical form.

Type-specific EM adaptations. Three question types use specialised rules for EM and its companion soft metrics. **(1) Judgement (FJ).** Only the graded EM above enters the main results; token-level F1, BLEU-1, and LLM-J are not applied to this type (recorded as 0, with no judge call) to avoid mixing scales with a non-textual answer form. **(2) Image-filename matching (FM whose gold parses to an img_<n> identifier).** Instead of the graded EM over $\mathcal{A}$, we use a binary image-id match against the canonical gold a_0 . Writing g for the normalised image id of a_0 and $\text{ids}(\hat{a})$ for the set of all `img_<n>` tokens extracted from the prediction,

$$\text{EM}_{\text{img}}(\hat{a}, a_0) = \begin{cases} 1 & g \in \text{ids}(\hat{a}), \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

This subtype computes no F1, BLEU-1, or LLM-J against a_0 (all recorded as 0), since the shared `img` prefix would otherwise inflate token overlap; set-based precision/recall/F1 over predicted and gold image ids are logged per item for diagnostics only and do not enter the main table. **(3) Text matching (FM with no parseable img_<n> in the gold).** EM scores 1.0 if $\text{norm}_{\text{em}}(\hat{a}) = \text{norm}_{\text{em}}(a_i)$ for any i and 0 otherwise (a full-string hit on any equivalent phrasing earns full credit, with no 2^{-i} discount). This subtype still computes F1, BLEU-1, and LLM-J against a_0 , but those soft scores are excluded from the TOTAL macro-average together with all FM and FJ items.

Token-Level F1. We use the SQuAD-style token-level F1 (Rajpurkar et al., 2016) between the prediction and the canonical gold a_0 . Let $P = \text{tok}(\hat{a})$ and $G = \text{tok}(a_0)$ be the predicted and gold token

multisets, and $|P \cap G|$ the number of overlapping tokens (with multiplicity). Then

$$\begin{aligned} \text{Prec} &= \frac{|P \cap G|}{|P|}, & \text{Rec} &= \frac{|P \cap G|}{|G|}, \\ \text{F1} &= \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}. \end{aligned} \quad (8)$$

When P and G are both empty we set $\text{F1} = 1$; when exactly one is empty we set $\text{F1} = 0$. F1 captures partial lexical overlap that strict EM ignores and is the standard metric for short-answer QA evaluation. In the main results, the TOTAL F1 is an equal-weight macro-average over the items remaining after **excluding all FM and FJ** questions.

BLEU-1. We report unigram BLEU with brevity penalty as introduced by Papineni et al. (2002), computed against the canonical gold a_0 . Let $c_{\text{clip}}(w) = \min(c_P(w), c_G(w))$ denote the clipped count of unigram w in the prediction, where c_P and c_G are the unigram counts in P and G . Then

$$\begin{aligned} \text{BP} &= \min(1, \exp(1 - |G|/|P|)), \\ \text{BLEU-1} &= \text{BP} \cdot \frac{\sum_{w \in P} c_{\text{clip}}(w)}{|P|}. \end{aligned} \quad (9)$$

Compared with F1, BLEU-1 imposes an additional penalty on overly short predictions through the brevity penalty BP; the empty-set boundary is handled as for F1. Its macro-average rule is the same as F1’s (TOTAL excludes FM and FJ).

LLM-as-a-Judge (LLM-J). Following Mem-Gallery (Bei et al., 2026), we use Qwen2.5-VL-32B-Instruct (Bai et al., 2025) as an impartial judge and ask it to score $\hat{a}$ against a_0 on a five-level rubric $\{0, 0.25, 0.5, 0.75, 1.0\}$ covering *incorrect*, *poor / tangential*, *partial / vague*, *good / minor imperfection*, and *correct / exact*. The verbatim judge prompt is reproduced in Appendix E. The judge may only emit one of the five levels; a parse failure is scored 0. LLM-J captures semantic equivalence that strict-token metrics miss (e.g., paraphrased correct answers or equivalent entity mentions). It is not applied to FJ or to the FM image subtype (no judge call, `llm_score` set to 0); the remaining types, including the FM text subtype, are still scored against a_0 , but the TOTAL LLM-J macro-averages only after **excluding all FM and FJ** items.

A graded LLM-as-a-Judge can in principle be sensitive to borderline correctness and answer phrasing. To check that our scoring is not biased by the choice of judge, we conduct a small-scale human

Judge	Text		Image		PDF	
	Count	Acc. (%)	Count	Acc. (%)	Count	Acc. (%)
Human Verification	50 / 50	100	50 / 50	100	50 / 50	100
Qwen2.5-VL-3B	52 / 48	96	50 / 50	96	48 / 52	96
DeepSeek-V3.2	49 / 51	95	51 / 49	97	50 / 50	98
Qwen2.5-VL-32B (Ours)	51 / 49	97	50 / 50	100	48 / 52	98

Table 3: Agreement of each judge with human verification on 50 sampled items per task type. “Count” reports the correct / incorrect split as scored by that judge; “Acc.” reports agreement (%) with human verification. Numbers are close across judges, indicating that LLM-J is stable across judge models.

verification on three reasoning-heavy task types — text-only, image-grounded, and PDF-grounded reasoning. For each task type we manually sample 50 correct answered and 50 wrong model generations together with their queries and gold answers, and re-score them with two alternative judges: a smaller in-family model (Qwen2.5-VL-3B-Instruct) and an out-of-family strong model (DeepSeek-V3.2). We then measure agreement with human verification on the same items. We binarize each judgment by treating scores of 0.75–1.0 as correct and 0–0.5 as incorrect, and report both the judgment counts and the resulting accuracy against the human-annotated sample.

Results are summarized in Table 3. The default judge (Qwen2.5-VL-32B-Instruct) achieves high agreement with human verification across all three task types, and both alternative judges produce closely matching accuracy. The consistency across judge families indicates that our LLM-J scores are not driven by an idiosyncratic preference of a single judge model, supporting the robustness of the LLM-as-a-Judge protocol used throughout the paper. Variance is further reduced in the main results (Table 4) by averaging over repeated runs with independent judging.

Overall Score. To rank systems with a single number, we report a composite *Overall Score* that combines the four answer-quality metrics,

$$\text{Overall} = 0.5 \text{ LLM-J} + 0.5 \cdot \frac{1}{3} (\text{EM} + \text{F1} + \text{BLEU-1}), \quad (10)$$

where each constituent is the per-type score macro-averaged over question types weighted by their item counts. The half weight on LLM-J, with the remaining half split equally among EM, F1, and BLEU-1, is deliberate: M³Exam answers are often compositional—bundling several elements such as multi-step procedures or cross-modal facts—rather than short spans, so the surface-matching metrics (EM, F1, BLEU-1) only partially credit a

correct-but-rephrased response. We therefore give the semantically-aware LLM-J the largest single weight, while still retaining the lexical metrics as a check against judge leniency. Only the Overall Score and the three component metrics EM, F1, and LLM-J appear in the main paper; BLEU-1 enters solely through this composite.

B.3 Baseline Descriptions

In this appendix section, we provide detailed descriptions of the advanced memory methods evaluated in Table 4. To enable a fair comparison between text-only and multimodal regimes, the text-only group is provided with high-quality image captions generated by Qwen2.5-VL-32B as a text-equivalent of the visual evidence the multimodal group sees.

Base (Full / Session, Text / Multimodal). The four *Base* configurations feed prior dialogue directly into the backbone without any retrieval or external memory module. *Base (Full)* concatenates the entire dialogue history, including all multimodal memory, into the context. To respect the backbone’s context-token limit, it estimates the token consumption of each image using a predefined per-image token cost and truncates the input once the running total would exceed the limit, so the longest histories are clipped rather than dropped. *Base (Session)* restricts the context to the single session in which the question is grounded; each is run in two regimes, where the text-only regime supplies captions of attached visuals while the multimodal regime supplies the original images (capped at ten per question). Comparing across the four configurations isolates the contribution of cross-session context (Full vs. Session) and of visual input (text vs. multimodal).

NaiveRAG (Lewis et al., 2020) encodes prior dialogue turn with a dense sentence encoder, indexes the resulting vectors in a FAISS store,

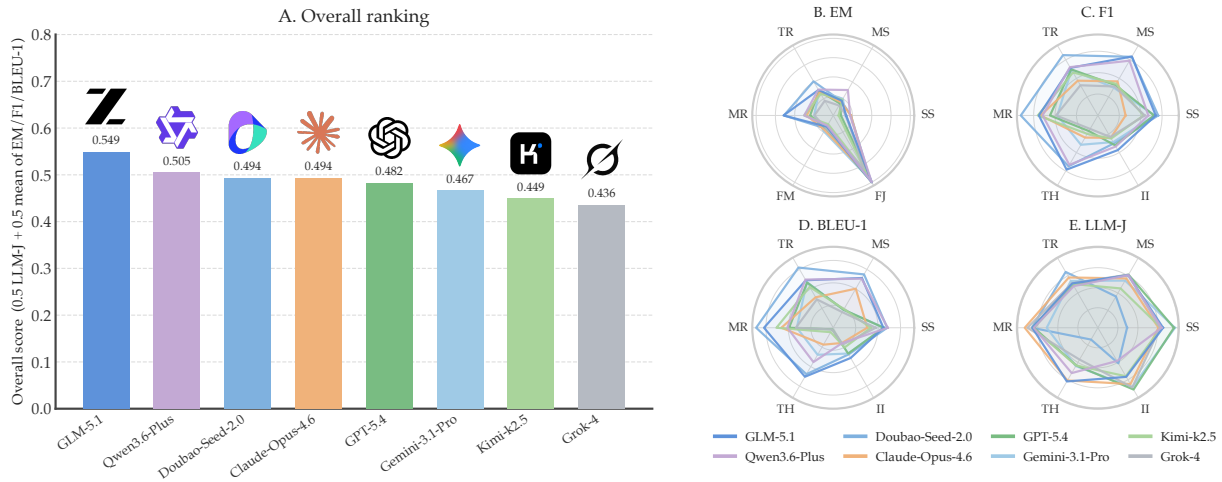


Figure 9: **Frontier-MLLM arena on M³Exam**. We evaluate eight frontier closed-source multimodal LLMs—GLM-5.1, Qwen3.6-Plus, Claude-Opus-4.6, Doubao-Seed-2.0, GPT-5.4, Gemini-3.1-Pro, Kimi-k2.5 and Grok-4—as the answering model on M³Exam, scoring every model under the four metrics (EM, F1, BLEU-1 and LLM-J). **Left**: overall ranking, where each model is scored under a unified standard combining the four metrics. **Right**: per-metric breakdown, one radar per metric, where each hexagon spans the six question types that the metric is defined on and each model is drawn as one colored polygon.

and at inference time retrieves the top- k chunks whose embeddings are nearest to the question. The retrieved chunks are concatenated into the model’s prompt without further reranking or summarization. <https://github.com/gusye1234/nano-graphrag>

A-Mem (Xu et al., 2025) is an agentic memory system for LLM agents that dynamically organises memories by drawing inspiration from the Zettelkasten method to build interconnected knowledge networks. When adding a new memory, A-Mem generates structured notes with contextual descriptions, keywords, and tags, then identifies connections with historical memories to establish links, while enabling memory evolution through updates to existing representations as new information is integrated. <https://github.com/WujiangXu/A-mem>

Mem0 / Mem0 (with vision). (Chhikara et al., 2025) is a production-style memory layer that extracts atomic facts from each dialogue turn, deduplicates them against the existing store via LLM-judged equivalence, and retrieves a top- K subset at inference time using dense semantic search; an optional graph variant adds typed entity relations between extracted facts. We evaluate two configurations: the text-only *Mem0* stores fact strings derived from captions of attached visuals, while *Mem0 (with vision)* additionally embeds the original images alongside the textual facts and retrieves both modalities through the same consolidation-and-recall

pipeline. <https://github.com/mem0ai/mem0>

MemoryOS. (Li et al., 2025) treats long-term memory as an operating system over a unified abstraction (*MemCube*) that wraps parametric, activation, and plaintext memory under one lifecycle manager covering allocation, eviction, migration, and access control. At inference time, queries are dispatched through a scheduler that surfaces the most relevant MemCubes across the three tiers; we use the plaintext-memory variant for fair comparison against the other text-based baselines. <https://github.com/MemTensor/MemOS>

UniversalRAG. (Yeo et al., 2025) maintains a modality- and granularity-aware corpus in which text passages, images, and clip-level chunks are each indexed separately, and trains a lightweight router that selects the modality and granularity most likely to contain the answer for a given query. Retrieval is then executed within the selected sub-index, and the top- $K=10$ results are concatenated into the prompt. <https://github.com/wgcyeo/UniversalRAG>

RAG-Anything. (Guo et al., 2025) is an all-in-one RAG framework that indexes heterogeneous content—text, images, tables, and equations—under a single shared embedding space and supports cross-modal top- $K=10$ retrieval through a unified query encoder. We use the framework’s default cross-modal configuration without task-specific tuning. <https://github.com/HKUDS/>

RAG-Anything

MIRIX. (Wang and Chen, 2025) is a multi-agent memory system that partitions long-term memory into six specialised stores—core, episodic, semantic, procedural, resource, and knowledge-vault—each managed by a dedicated agent that decides whether incoming content belongs to its store. At inference time, a router agent dispatches the query to the relevant store(s) and aggregates the retrieved items before prompting the backbone. <https://github.com/Mirix-AI/MIRIX>

MemVerse. (Liu et al., 2025) is a multimodal memory layer for lifelong-learning agents that maintains visual and textual memories under explicit lifecycle operations—store, recall, update, and forget—each implemented as an LLM-driven policy over an embedding-indexed multimodal store. New observations trigger update or store operations, while retrieval at inference time issues a recall over the indexed embeddings. <https://github.com/KnowledgeXLab/MemVerse>

NGM (Neural Graph Memory). (Fisher, 2025) represents long-term memory as a heterogeneous graph in which nodes correspond to dialogue facts and visual entities and edges encode co-occurrence, temporal, and cross-modal relations. New turns are converted into subgraphs that are merged into the global graph; retrieval performs neighbourhood expansion from query-anchor nodes and returns the induced subgraph as context. <https://github.com/StuckInTheNet/Neural-Graph-Memory-NGM>

C Extended Experiments

C.1 Alternative MLLM Backbone

For space, the main-paper table reports a condensed view of the results—two metrics per type and a single composite score. Here we provide the *complete* per-type, four-metric breakdown (EM, F1, BLEU-1, and LLM-J) for every memory system, and we further repeat the full evaluation across *four answering backbones* of different families and scales to confirm that our findings are not an artifact of one backbone: two open-source MLLMs (Qwen2.5-VL-7B, Table 4; and the smaller Qwen2.5-VL-3B, Table 5) and two strong proprietary models (GPT-5-Nano, Table 6; and Doubao-Seed-2.0-Pro, Table 7). All agentic-memory systems are deployed identically on each backbone, so differences within a

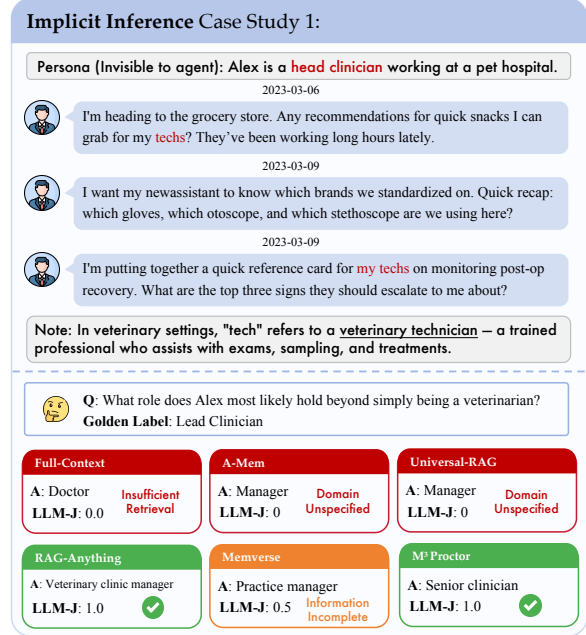


Figure 10: Case study with Implicit Inference Example 1 (Alex, Veterinarian).

table reflect memory design and differences across tables reflect the backbone.

Findings are consistent across backbones. The conclusions drawn in the main paper hold on all four backbones. **(1) Memory is essential.** The memory-less *Full Context* baseline sits far below every memory system on each backbone (overall LLM-J of only 0.14–0.33), confirming that raw long-context access cannot substitute for structured retrieval. **(2) Multimodal memory beats text-only memory.** On every backbone the multimodal systems occupy the top of the table while the text-only group trails (e.g., on Qwen2.5-VL-7B the best text-only system reaches 0.44 LLM-J versus 0.53–0.58 for the multimodal group), reaffirming that M³Exam’s questions genuinely require visual evidence. **(3) M³Proctor leads regardless of backbone.** M³Proctor attains the best overall LLM-J on all four backbones—0.582 (Qwen2.5-VL-7B), 0.476 (Qwen2.5-VL-3B), 0.597 (GPT-5-Nano), and 0.552 (Doubao-Seed-2.0-Pro)—ahead of the strongest prior system (MIRIX) in every case, with the gain again concentrated on the cross-modal and implicit-intent types. Backbone scale shifts the absolute numbers (the 3B backbone lowers all systems, and the proprietary backbones raise the whole field) but never the ordering: M³Proctor on the modest 3B backbone (0.476) still outperforms most baselines run on the larger 7B backbone, indicating that its advantage

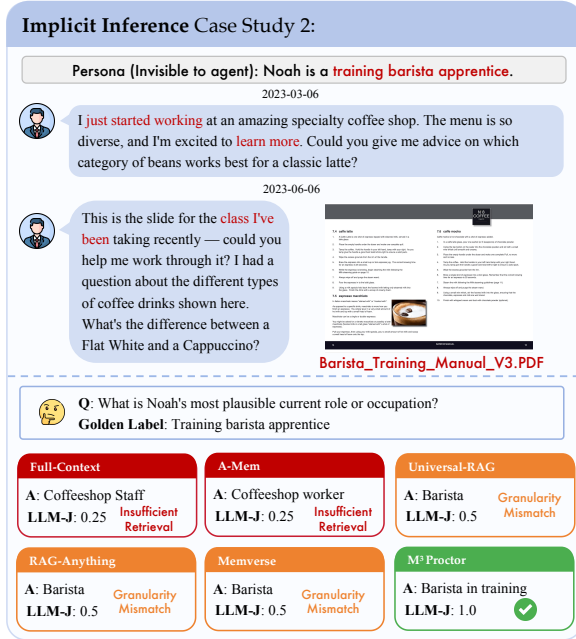


Figure 11: Case study with Implicit Inference Example 2 (Noah, Barista Apprentice).

stems from modality-aware evidence management rather than backbone capacity.

We further ask how far the current frontier of closed-source MLLMs is from solving M³Exam on its own. The main paper already reports five such models alongside the memory systems; here we broaden the panel into a head-to-head arena of *eight* leading models—GLM-5.1, Qwen3.6-Plus, Claude-Opus-4.6, Doubao-Seed-2.0, GPT-5.4, Gemini-3.1-Pro, Kimi-k2.5, and Grok-4—each used as the answering model and scored under all four metrics (Figure 9). To isolate *reasoning* ability from long-context handling, we do not run these models over the full conversation history. Instead, for each question we use its annotated supporting facts to locate the exact session(s)—together with their attached images and PDF pages—that the answer depends on, and provide only that evidence to the model. This oracle-context setting removes the retrieval burden entirely and upper-bounds what each model could achieve given perfect memory, so any remaining error reflects a reasoning or grounding limitation rather than a failure to recall. To summarise each model with a single indicator, we additionally report a unified score that weights semantic correctness against lexical overlap: 0.5 LLM-J + 0.5 mean(EM, F1, BLEU-1). Because many answers in M³Exam are compositional—bundling several elements such as multi-step procedures or cross-modal facts—rather than short spans,

the surface-matching metrics (EM, F1, BLEU-1) only partially credit a correct-but-rephrased response; we therefore give the semantically-aware LLM-J the largest single weight (0.5) and let the three lexical metrics share the other half.

Even frontier MLLMs are far from solving M³Exam. Under this favourable setting, no model exceeds an overall score of 0.55: the eight models cluster tightly between 0.445 and 0.550 (Figure 9A), with GLM-5.1 narrowly on top and the remaining models—including the latest GPT, Gemini, Claude, and Grok—separated by only a few points. The arena is also not won by a single model: the per-metric radars (Figure 9B–E) show different leaders on different axes, and every model follows the same uneven profile—relatively strong on single-session recall (ss) and factual judgement (FJ) but dropping sharply on multimodal reasoning (MR) and the interpreting types (thematic TH and implicit inference II), where the answer must be composed across modalities or inferred from unstated context. Two observations follow. First, even when the gold evidence is handed to the model directly, frontier MLLMs still fail on a large fraction of items, exposing genuine deficits in cross-modal grounding and implicit-intent reasoning rather than mere context-length limits; the consistency of this weakness across eight independent models shows it is a property of the task, not of any one system. Second, the oracle-context score is an *upper bound* that a deployed agent without access to gold supporting facts cannot reach; closing the gap to it requires a memory system that surfaces the right multimodal evidence on its own, which motivates the retrieval-augmented methods studied in the main paper.

C.2 Full Context Evaluation

The main paper evaluates each backbone in a *session-only* setting: for every question we feed only the gold supporting session(s), which relieves the model of retrieval and lets us measure its raw answering ability in isolation. A natural question is whether this simplification is necessary—could a backbone simply ingest the *full* conversation and find the evidence itself? To answer this, we re-run the same backbones with the entire history in context and compare the two regimes in the multimodal setting (Figure 12), where the gap reported is *Session-only–Full context*.

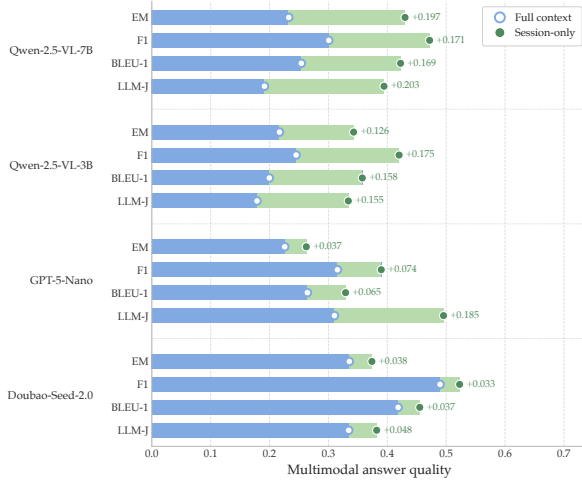


Figure 12: **The full history adds context and retrieval pressure.** For each backbone and metric in the multimodal regime, we compare feeding only the gold supporting session(s) (*Session-only*, retrieval relieved) against feeding the entire conversation (*Full context*). Bars are overlapped on a shared baseline; the green tail and its label give the *Session-only-Full context* gap, i.e. the accuracy lost to processing the full history.

Feeding the full history imposes both context and retrieval pressure. All sixteen comparisons (four backbones $\times$ four metrics) favour the session-only setting: handing a model the whole conversation *lowers* answer quality in every case, because it must now both withstand a much longer context and locate the relevant rounds amid many distractors. The penalty is largest on the open Qwen-2.5-VL backbones—up to +0.197/+0.203 EM/LLM-J on the 7B model and +0.126 to +0.175 across metrics on the 3B model, around a fifth of their score. The proprietary models tolerate the longer input better but still degrade throughout: Doubao-Seed-2.0 by +0.033 to +0.048, and GPT-5-Nano by only +0.037 to +0.074 on the lexical metrics yet +0.185 on LLM-J, showing the extra context chiefly eroded the *semantic* quality of its answers. Because even frontier MLLMs lose accuracy once retrieval and context pressure are added back, the full-history regime is not a viable substitute for structured memory—which is precisely why the main paper isolates answering ability with the session-only setting and why a memory that surfaces the right evidence on its own is needed in deployment.

C.3 Case Study: Implicit Inference

Figures 10 and 11 illustrate why implicit-inference questions expose the limits of current systems. In both, the user’s true role is never stated and must

be composed from scattered cues—Alex repeatedly instructing his *techs* and standardizing clinic equipment (Figure 10), or Noah describing himself as newly hired and studying a training manual (Figure 11). Systems fail in distinct ways. Memory-less and text-only baselines (Full-Context, A-Mem) retrieve too little context and answer generically (“Doctor”, “Manager”, “Coffeeshop staff”), while multimodal baselines that retrieve more evidence still misjudge the granularity of the inference, over- or under-specifying the role (“Veterinary clinic manager”, plain “Barista”). M³Proctor, in contrast, both surfaces the dispersed supporting evidence and calibrates the inference to the right level (“Senior clinician”, “Barista in training”), matching the gold label. These cases show that answering correctly requires not just recalling evidence but interpreting unstated, domain-grounded intent that prior systems systematically miss.

D Persona and Core-Event Specifications

Each persona scenario in M³Exam is seeded from a *persona* (who the user is and what their interactions cover) and a *core event* (the storyline that the multi-session timeline elaborates). Table 8 summarises both seeds for all fifteen subsets; these are the inputs to the timeline-construction prompts described in Section E.

Table 4: QA performance of M³Exam on **Qwen-2.5-VL-7B**. For each metric, the best and second-best scores are marked in **bold** and with underline, respectively.

 Qwen-2.5-VL-7B		SS	MS	TR	MR	FM	FJ	TH	II	Avg.
<i>Text-Only</i>										
Full Context (without memory)	EM	0.1638	0.1148	0.1865	0.1067	0.0596	0.5189	–	–	0.1671
	F1	0.3342	0.2598	0.2947	0.1801	–	–	0.1458	0.1572	0.2370
	BLEU-1	0.2898	0.1885	0.2752	0.1603	–	–	0.0677	0.1112	0.1964
	LLM-J	0.1938	0.1592	0.2407	0.1229	–	–	0.0123	0.0263	0.1443
NaiveRAG	EM	0.2507	0.1849	0.3613	0.1522	0.2569	0.7840	–	–	0.2914
	F1	0.4430	0.3117	0.2349	0.2080	–	–	0.1321	0.1263	0.2572
	BLEU-1	0.3945	0.2323	0.2151	0.1815	–	–	0.0373	0.0798	0.2107
	LLM-J	0.6547	0.4429	0.4167	0.2572	–	–	0.2806	0.3202	0.3967
A-Mem	EM	0.3347	0.2887	0.1532	0.1844	0.2875	0.9243	–	–	0.3100
	F1	0.5921	0.3998	0.1298	0.2403	–	–	0.1361	0.1227	0.2867
	BLEU-1	0.5246	0.2996	0.1255	0.2145	–	–	0.0321	0.0509	0.2339
	LLM-J	<u>0.7247</u>	0.5345	0.1697	0.3727	–	–	0.3493	0.2872	0.4132
Mem0 (w/o vision)	EM	0.2947	0.2950	0.1893	0.1765	0.1544	0.9243	–	–	0.2886
	F1	0.5343	0.3837	0.1361	0.2471	–	–	0.1346	0.0962	0.2746
	BLEU-1	0.4735	0.2895	0.1341	0.2193	–	–	0.0310	0.0452	0.2254
	LLM-J	0.6957	0.5448	0.1853	0.3787	–	–	0.3843	0.3000	0.4191
MemoryOS	EM	0.3027	0.2414	0.1453	0.1508	0.3150	<u>0.9733</u>	–	–	0.2970
	F1	0.5410	0.3399	0.1248	0.2146	–	–	0.1274	0.1162	0.2587
	BLEU-1	0.4779	0.2522	0.1235	0.1855	–	–	0.0281	0.0459	0.2088
	LLM-J	0.7047	0.5145	0.3977	0.3299	–	–	0.3327	0.3175	0.4392
<i>Multimodality</i>										
Full Context (without memory)	EM	0.1587	0.1372	0.2258	0.2182	0.1361	0.6726	–	–	0.2326
	F1	0.3703	0.3247	0.3375	0.2982	–	–	0.1825	0.1752	0.3006
	BLEU-1	0.3122	0.2326	0.3201	0.2774	–	–	0.1018	0.1254	0.2539
	LLM-J	0.1976	0.1958	0.2951	0.2255	–	–	0.0211	0.0294	0.1913
Universal -RAG	EM	0.2914	0.1777	0.3002	0.2035	0.5367	0.8241	–	–	0.3457
	F1	0.5858	0.4349	0.3903	0.2938	–	–	0.2691	0.1586	0.3745
	BLEU-1	0.5174	0.3165	0.3798	0.2621	–	–	0.1517	0.0894	0.3146
	LLM-J	0.7328	0.5769	0.4431	0.4369	–	–	<u>0.4872</u>	0.5559	0.5283
RAG -Anything	EM	0.2977	0.1308	0.3597	0.2844	0.4572	0.8508	–	–	0.3630
	F1	0.5849	0.3940	0.4353	0.3863	–	–	0.2782	0.1873	0.4063
	BLEU-1	0.5137	0.2881	<u>0.4193</u>	0.3562	–	–	<u>0.1673</u>	<u>0.1041</u>	0.3468
	LLM-J	0.6418	0.5474	<u>0.4465</u>	0.4970	–	–	0.4390	<u>0.6201</u>	<u>0.5241</u>
Mem0 (with vision)	EM	0.2547	0.2727	0.2027	0.3332	0.1942	0.9176	–	–	0.3272
	F1	0.4798	0.4181	0.1511	0.4459	–	–	0.1485	0.1286	0.3327
	BLEU-1	0.4288	<u>0.3058</u>	0.1473	0.4140	–	–	0.0356	0.0548	0.2783
	LLM-J	0.6457	0.5548	0.1837	0.5607	–	–	0.3527	0.5973	0.4811
MemVerse	EM	0.2451	0.1411	0.2606	0.3589	0.5474	0.9042	–	–	0.3760
	F1	0.4850	0.3895	0.3229	0.4591	–	–	0.2518	0.1849	0.3826
	BLEU-1	0.4243	0.2805	0.3146	0.4293	–	–	0.1485	0.0933	0.3265
	LLM-J	0.6406	0.5662	0.4096	<u>0.5613</u>	–	–	0.4223	0.6048	0.5345
NGM	EM	0.1808	0.1448	0.1198	0.3894	0.6208	0.9042	–	–	0.3591
	F1	0.3559	0.3453	0.1716	0.4708	–	–	0.2288	0.1807	0.3231
	BLEU-1	0.3121	0.2496	0.1650	0.4511	–	–	0.1398	0.0996	0.2777
	LLM-J	0.5266	0.4660	0.1601	0.5479	–	–	0.3905	0.6060	0.4427
MIRIX	EM	0.2959	0.0688	0.3980	0.2900	0.3807	0.8664	–	–	0.3527
	F1	0.4954	0.2853	0.4564	0.4076	–	–	0.2999	0.1054	0.3800
	BLEU-1	0.4467	0.1486	0.4520	0.3694	–	–	0.2572	0.0481	0.3299
	LLM-J	0.6378	0.4875	0.4949	<u>0.5775</u>	–	–	0.5917	0.5375	<u>0.5578</u>
M³Proctor (Our Method)	EM	0.2541	0.1395	0.4042	0.3572	<u>0.5688</u>	0.9844	–	–	0.4140
	F1	0.4993	0.3697	0.3952	0.4305	–	–	0.2202	0.3021	0.3935
	BLEU-1	0.4378	0.2466	0.4101	0.4125	–	–	0.1235	0.2823	0.3491
	LLM-J	0.6924	<u>0.5667</u>	<u>0.4877</u>	0.6063	–	–	0.4786	0.6515	0.5820

Table 5: QA performance of M³Exam on **Qwen-2.5-VL-3B**. For each metric, the best and second-best scores are marked in **bold** and with underline, respectively.

 Qwen-2.5-VL-3B		SS	MS	TR	MR	FM	FJ	TH	II	Avg.
<i>Text-Only</i>										
Full Context (without memory)	EM	0.1612	0.1567	0.1118	0.1287	0.0535	0.6904	–	–	0.1814
	F1	0.2408	0.2812	0.2118	0.1862	–	–	0.1388	0.1313	0.2058
	BLEU-1	0.2185	0.2387	0.2001	0.1662	–	–	0.0605	0.0822	0.1748
	LLM-J	0.1755	0.1747	0.1958	0.1338	–	–	0.0347	0.0188	0.1396
NaiveRAG	EM	0.2667	0.2027	0.3773	0.2494	0.3394	0.8976	–	–	0.3488
	F1	0.3657	0.3318	0.2183	0.2961	–	–	0.1205	0.0859	0.2632
	BLEU-1	0.3320	0.2252	0.2070	0.2738	–	–	0.0244	0.0439	0.2184
	LLM-J	0.5509	0.3693	0.4274	0.4342	–	–	0.2380	0.3331	0.4147
A-Mem	EM	0.3947	0.2977	0.2373	0.2133	0.3563	0.9109	–	–	0.3530
	F1	0.5585	0.4221	0.1531	0.2691	–	–	0.1293	0.0729	0.2919
	BLEU-1	0.5012	0.3063	0.1483	0.2345	–	–	0.0322	0.0227	0.2385
	LLM-J	0.6183	0.4282	0.2484	0.4075	–	–	0.2839	0.3331	0.3988
Mem0 (w/o vision)	EM	0.3867	0.2677	0.2027	0.2133	0.3150	0.9443	–	–	0.3386
	F1	0.5394	0.4147	0.1666	0.2780	–	–	0.1450	0.0899	0.2955
	BLEU-1	<u>0.4820</u>	<u>0.3001</u>	0.1623	0.2406	–	–	0.0394	0.0407	0.2407
	LLM-J	<u>0.6215</u>	0.4176	0.2503	0.3915	–	–	0.2881	0.3554	0.3959
MemoryOS	EM	0.3787	0.2723	0.2507	0.1934	0.2783	0.8508	–	–	0.3262
	F1	0.4919	0.4036	0.1417	0.2492	–	–	0.1242	0.0606	0.2675
	BLEU-1	0.4393	0.2934	0.1427	0.2205	–	–	0.0249	0.0273	0.2197
	LLM-J	0.6094	0.4061	0.2472	0.3793	–	–	0.2712	0.3126	0.3827
<i>Multimodality</i>										
Full Context (without memory)	EM	0.1294	0.1108	0.1188	0.1662	0.1453	0.9087	–	–	0.2168
	F1	0.2811	0.2557	0.2226	0.3003	–	–	0.1768	0.0916	0.2451
	BLEU-1	0.2296	0.1959	0.1969	0.2658	–	–	0.0967	0.0491	0.1996
	LLM-J	0.1960	0.1527	0.2308	0.2593	–	–	0.0138	0.0028	0.1787
Universal -RAG	EM	0.1122	0.0250	0.3673	0.2167	0.3807	0.9243	–	–	0.2966
	F1	0.3375	0.2742	0.4855	0.3263	–	–	0.1319	0.1537	0.3163
	BLEU-1	0.2718	0.1506	<u>0.4702</u>	0.2817	–	–	0.0530	0.0622	0.2543
	LLM-J	0.6047	0.3319	0.4940	0.4395	–	–	0.3088	0.3079	0.4397
RAG -Anything	EM	0.1224	0.0375	0.4107	0.2750	0.4404	0.8664	–	–	0.3260
	F1	0.3539	0.2596	0.4963	0.4118	–	–	0.1198	0.1517	0.3418
	BLEU-1	0.2839	0.1451	0.4889	0.3601	–	–	0.0374	0.0757	0.2807
	LLM-J	0.5876	0.3378	<u>0.5196</u>	0.4877	–	–	0.2672	0.3362	0.4534
Mem0 (with vision)	EM	0.2227	0.2227	0.1933	0.4054	0.2630	0.9131	–	–	0.3418
	F1	0.4191	0.3807	0.1604	0.4781	–	–	0.1523	0.0887	0.3242
	BLEU-1	0.3605	0.3143	0.1506	<u>0.4475</u>	–	–	0.0416	0.0363	0.2764
	LLM-J	0.5886	<u>0.4253</u>	0.2482	<u>0.5497</u>	–	–	0.3086	<u>0.3420</u>	0.4365
MemVerse	EM	0.1437	0.1157	0.2218	0.2634	0.4541	0.8330	–	–	0.3021
	F1	0.3615	0.3464	0.3058	0.3698	–	–	0.2439	0.1313	0.3199
	BLEU-1	0.3002	0.2506	0.2968	0.3382	–	–	<u>0.1436</u>	0.0629	0.2671
	LLM-J	0.5427	0.3991	0.3805	0.5083	–	–	<u>0.3513</u>	0.3096	0.4406
NGM	EM	0.2048	0.1330	0.0937	0.3584	0.4312	0.8552	–	–	0.3155
	F1	0.3642	0.3269	0.1618	0.4855	–	–	0.2028	0.1787	0.3210
	BLEU-1	0.3184	0.2311	0.1577	0.4523	–	–	0.1008	0.0861	0.2695
	LLM-J	0.5065	0.3666	0.2329	0.5306	–	–	0.2950	<u>0.3249</u>	0.4014
MIRIX	EM	0.2143	0.0250	0.3316	0.1000	<u>0.4511</u>	0.8463	–	–	0.2797
	F1	0.4138	0.2408	0.4556	0.2610	–	–	0.2624	0.0844	0.3109
	BLEU-1	0.3594	0.1371	0.4348	0.2169	–	–	0.2125	0.0391	0.2599
	LLM-J	0.5105	0.3081	0.4994	0.5716	–	–	0.3828	0.2568	<u>0.4618</u>
M³Proctor (Our Method)	EM	0.3000	0.1597	0.3503	0.2175	0.4450	<u>0.9310</u>	–	–	0.3546
	F1	0.5218	0.2876	0.4790	0.4087	–	–	0.1913	0.1753	0.3828
	BLEU-1	0.4672	0.1920	0.4262	0.3267	–	–	0.1776	0.1156	0.3190
	LLM-J	0.6450	0.3425	0.5410	0.4975	–	–	0.3128	0.3305	0.4764

Table 6: QA performance of M³Exam on **GPT-5-Nano**. For each metric, the best and second-best scores are marked in **bold** and with underline, respectively.

 GPT-5-Nano		SS	MS	TR	MR	FM	FJ	TH	II	Avg.
<i>Text-Only</i>										
Full Context (without memory)	EM	0.1133	0.0973	0.1680	0.1774	0.0902	0.9644	–	–	0.2214
	F1	0.3896	0.4041	0.2982	0.3083	–	–	0.2636	0.1763	0.3205
	BLEU-1	0.3194	0.2929	0.2814	0.2675	–	–	0.1585	0.1376	0.2613
	LLM-J	0.4257	0.3713	0.2184	0.2432	–	–	0.3434	0.4037	0.3147
NaiveRAG	EM	0.2683	0.2042	0.3787	0.2478	0.5000	0.8953	–	–	0.3728
	F1	0.4082	0.3303	0.2197	0.2978	–	–	0.1218	0.0844	0.2716
	BLEU-1	0.3728	0.2238	0.2085	0.2723	–	–	0.0258	0.0426	0.2257
	LLM-J	0.5982	0.4012	0.4691	0.4738	–	–	0.2625	0.3613	0.4524
A-Mem	EM	0.3938	0.2965	0.2392	0.2152	0.4786	0.9109	–	–	0.3718
	F1	0.4213	0.3205	0.1543	0.2705	–	–	0.1278	0.0742	0.2520
	BLEU-1	0.4567	0.2082	0.1495	0.2358	–	–	0.0307	0.0241	0.2163
	LLM-J	0.6695	0.4673	0.2715	0.4422	–	–	0.3078	0.3565	0.4329
Mem0 (w/o vision)	EM	0.3852	0.2692	0.2042	0.2147	0.4037	0.9421	–	–	0.3522
	F1	0.6038	0.4163	0.1681	0.2766	–	–	0.1432	0.0883	0.3072
	BLEU-1	0.5373	0.2986	0.1638	0.2391	–	–	0.0408	0.0421	0.2508
	LLM-J	0.6732	0.4533	0.2762	0.4255	–	–	0.3128	0.3838	0.4303
MemoryOS	EM	0.3771	0.2740	0.2492	0.1952	0.2798	0.8530	–	–	0.3268
	F1	0.5473	0.4019	0.1429	0.2477	–	–	0.1226	0.0623	0.2773
	BLEU-1	0.4884	0.2947	0.1442	0.2192	–	–	0.0233	0.0258	0.2285
	LLM-J	0.6602	0.4751	0.2632	0.4076	–	–	0.2945	0.3411	0.4183
<i>Multimodality</i>										
Full Context (without memory)	EM	0.0838	0.0973	0.1670	0.2200	0.1514	0.8597	–	–	0.2257
	F1	0.3123	0.3440	0.3328	0.3438	–	–	0.2259	0.2503	0.3153
	BLEU-1	0.2553	0.2580	0.3147	0.3037	–	–	0.1388	<u>0.2110</u>	0.2642
	LLM-J	0.3707	0.3146	0.2187	0.3148	–	–	0.3084	0.3752	0.3102
Universal -RAG	EM	0.1633	0.0750	0.3776	0.2333	0.2798	0.9042	–	–	0.3013
	F1	0.4202	0.3465	0.4112	0.3989	–	–	0.1554	0.2504	0.3583
	BLEU-1	0.3595	0.1875	0.4612	0.3531	–	–	0.0330	0.1936	0.3020
	LLM-J	<u>0.6939</u>	0.6375	0.4898	0.5750	–	–	0.4250	0.8250	<u>0.5913</u>
RAG -Anything	EM	0.1122	0.0625	0.3673	0.2056	0.3593	0.9844	–	–	0.3018
	F1	0.4126	0.3204	0.3299	0.3800	–	–	0.1407	0.2035	0.3267
	BLEU-1	0.3406	0.1875	0.3724	0.3248	–	–	0.0152	0.1504	0.2679
	LLM-J	0.6721	0.5290	0.4809	0.5343	–	–	0.3986	0.5980	0.5378
Mem0 (with vision)	EM	0.2208	0.2243	0.1948	0.4072	0.0612	0.9265	–	–	0.3137
	F1	0.4178	0.3793	0.1622	0.4767	–	–	0.1508	0.0875	0.3235
	BLEU-1	0.3623	0.2858	0.1493	<u>0.4462</u>	–	–	0.0432	0.0348	0.2719
	LLM-J	0.6212	0.4488	0.2562	0.5738	–	–	0.3258	0.3637	0.4580
MemVerse	EM	0.0816	0.0250	0.4617	0.2333	0.4801	0.9265	–	–	0.3277
	F1	0.2602	0.2541	0.3875	0.4307	–	–	0.0995	0.2340	0.3118
	BLEU-1	0.2185	0.1459	0.3997	0.3707	–	–	0.1151	0.1740	0.2707
	LLM-J	0.6224	0.5312	<u>0.5255</u>	<u>0.6167</u>	–	–	0.3917	<u>0.7375</u>	0.5706
NGM	EM	0.1020	0.1225	0.1224	0.3667	0.3394	0.9443	–	–	0.2992
	F1	0.2824	0.2896	0.1959	<u>0.4283</u>	–	–	0.0666	0.2158	0.2781
	BLEU-1	0.2354	0.1651	0.1880	0.3830	–	–	0.0152	0.1559	0.2262
	LLM-J	0.4898	0.5188	0.1633	0.5889	–	–	0.1917	0.7000	0.4402
MIRIX	EM	0.1224	0.1043	0.4031	0.3281	<u>0.5245</u>	0.9666	–	–	0.3705
	F1	0.4374	0.2888	<u>0.4032</u>	0.4993	–	–	0.2929	0.2305	0.3940
	BLEU-1	0.3671	0.1621	0.3665	0.4481	–	–	0.2283	0.1576	0.3277
	LLM-J	0.6342	0.5208	0.4004	0.6497	–	–	<u>0.5119</u>	0.6385	0.5626
M³Proctor (Our Method)	EM	0.1800	0.2307	0.2307	0.3394	0.5306	<u>0.9777</u>	–	–	0.3717
	F1	0.5292	0.3490	0.4147	0.3174	–	–	0.2155	0.2270	0.3617
	BLEU-1	0.4537	0.2841	0.3710	0.2279	–	–	0.1623	0.2516	0.3001
	LLM-J	0.7317	<u>0.5500</u>	0.5350	0.5975	–	–	0.5500	0.5875	0.5969

Table 7: QA performance of M³Exam on **Doubao-Seed-2.0-Pro**. For each metric, the best and second-best scores are marked in **bold** and with underline, respectively.

 Doubao-Seed-2.0-Pro		SS	MS	TR	MR	FM	FJ	TH	II	Avg.
<i>Text-Only</i>										
Full Context (without memory)	EM	0.1211	0.1400	0.3925	0.2040	0.0979	0.9137	–	–	0.2718
	F1	<u>0.4871</u>	<u>0.4459</u>	0.5253	0.3094	–	–	<u>0.3313</u>	<u>0.3289</u>	0.4085
	BLEU-1	0.3961	0.3373	<u>0.5028</u>	0.2790	–	–	<u>0.2123</u>	<u>0.2824</u>	0.3454
	LLM-J	0.1977	0.1912	<u>0.4929</u>	0.2560	–	–	<u>0.0391</u>	<u>0.0593</u>	0.2420
NaiveRAG	EM	0.2626	0.2008	0.3777	0.2463	0.4954	0.9020	–	–	0.3708
	F1	<u>0.4069</u>	<u>0.3325</u>	0.2181	0.2956	–	–	0.1168	0.0842	0.2701
	BLEU-1	0.3695	0.2257	0.2059	0.2714	–	–	0.0235	0.0415	0.2243
	LLM-J	0.5940	0.4014	0.4621	0.4753	–	–	0.2606	0.3645	0.4507
A-Mem	EM	0.2488	0.2772	0.1187	0.1157	0.2018	0.8953	–	–	0.2539
	F1	0.3363	<u>0.3837</u>	0.0830	0.1559	–	–	0.0980	0.1273	0.2001
	BLEU-1	0.3108	0.3087	0.0756	0.1411	–	–	0.0284	0.0916	0.1680
	LLM-J	0.4448	0.4837	0.1250	0.2636	–	–	0.1694	0.4890	0.3090
Mem0 (w/o vision)	EM	<u>0.2626</u>	0.2676	0.1529	0.1847	0.5734	0.9488	–	–	0.3400
	F1	<u>0.4829</u>	0.4368	0.1430	0.2671	–	–	0.1788	0.1538	0.2894
	BLEU-1	0.4177	<u>0.3422</u>	0.1302	0.2303	–	–	0.0438	0.1268	0.2329
	LLM-J	0.6503	0.5839	0.2339	0.4337	–	–	0.3491	0.5048	0.4528
MemoryOS	EM	0.2056	0.2876	0.1901	0.1713	0.3135	0.8953	–	–	0.2919
	F1	0.4422	0.4142	0.1354	0.2723	–	–	0.1439	0.1644	0.2753
	BLEU-1	0.3761	0.3076	0.1250	0.2331	–	–	0.0244	0.1305	0.2179
	LLM-J	0.6326	<u>0.5537</u>	0.1848	0.4358	–	–	0.3025	0.5158	0.4316
<i>Multimodality</i>										
Full Context (without memory)	EM	0.1220	0.1311	0.2384	0.4994	0.1575	0.9777	–	–	0.3355
	F1	0.4869	0.4700	0.4491	0.6334	–	–	0.3384	0.3311	0.4896
	BLEU-1	0.4051	0.3628	0.4039	0.5876	–	–	0.2127	0.2762	0.4183
	LLM-J	0.2513	0.2181	0.4498	0.5561	–	–	0.0390	0.0893	0.3344
Universal -RAG	EM	0.1020	0.0500	0.4923	0.2444	0.4373	0.9510	–	–	0.3392
	F1	0.3816	0.3400	0.5091	0.3921	–	–	0.1493	0.2502	0.3666
	BLEU-1	0.3082	0.2179	0.5057	0.3452	–	–	0.0260	0.2075	0.3037
	LLM-J	0.6362	0.5511	0.5015	0.5732	–	–	0.2978	0.5958	0.5377
RAG -Anything	EM	0.0816	0.0500	0.4337	0.2444	0.4404	<u>0.9666</u>	–	–	0.3272
	F1	0.3605	0.2947	0.4738	0.3729	–	–	0.1618	0.2862	0.3479
	BLEU-1	0.2855	0.2007	0.4562	0.3304	–	–	0.0355	0.2532	0.2877
	LLM-J	0.6047	0.4874	0.5061	0.5395	–	–	0.3090	<u>0.6054</u>	0.5159
Mem0 (with vision)	EM	0.2186	0.2185	0.1964	0.4012	0.3609	0.9287	–	–	0.3562
	F1	0.4194	0.3833	0.1553	0.4780	–	–	0.1473	0.1868	0.3305
	BLEU-1	0.3596	0.2894	0.1454	0.4441	–	–	0.0420	0.1339	0.2779
	LLM-J	0.6198	0.4522	0.2521	0.5740	–	–	0.3228	0.3652	0.4572
MemVerse	EM	0.0816	0.0375	0.4439	0.4000	<u>0.5994</u>	0.9577	–	–	0.3907
	F1	0.3062	0.2543	0.3946	0.4444	–	–	0.1544	0.2700	0.3346
	BLEU-1	0.2438	0.1811	0.3855	0.4017	–	–	0.0354	0.2327	0.2819
	LLM-J	0.5979	0.4902	0.4366	0.6587	–	–	0.3318	0.5865	0.5363
NGM	EM	0.0204	0.0375	0.1531	0.4222	0.5612	0.9131	–	–	0.3235
	F1	0.1865	0.2596	0.1520	<u>0.5887</u>	–	–	0.0941	0.2864	0.3011
	BLEU-1	0.1457	0.1553	0.1469	<u>0.5431</u>	–	–	0.0112	0.2363	0.2509
	LLM-J	0.3827	0.4938	0.1633	0.7472	–	–	0.1250	0.6500	0.4500
MIRIX	EM	0.3265	0.0250	0.2449	0.1778	0.3777	0.9532	–	–	0.3036
	F1	0.5475	0.2780	0.3562	0.3427	–	–	0.2551	0.1794	0.3512
	BLEU-1	0.4921	0.1707	0.3344	0.2927	–	–	0.2031	0.1475	0.2983
	LLM-J	<u>0.6735</u>	0.3750	0.5255	<u>0.6861</u>	–	–	0.4250	0.3125	<u>0.5482</u>
M³Proctor (Our Method)	EM	0.1838	0.1893	0.3860	0.3973	0.6743	<u>0.9666</u>	–	–	0.4300
	F1	0.4436	0.2171	0.4525	0.4657	–	–	0.2272	0.3039	0.3826
	BLEU-1	0.3698	0.2574	0.4353	0.3970	–	–	0.1940	0.2927	0.3476
	LLM-J	0.7850	0.4562	<u>0.5100</u>	0.5775	–	–	<u>0.3583</u>	0.4750	0.5519

Table 8: Persona and core-event seeds for the fifteen subsets of M³Exam. The persona defines the user’s identity and the scope of their interactions; the core event defines the storyline that the multi-session timeline elaborates.

Subset	Persona	Core Event
Alex (Veterinarian)	Head clinician of a pet hospital handling a broad caseload of dogs, cats, and exotics; takes his work seriously, documents cases photographically, teaches techniques to his techs, and is a caring cat owner and a big snacks fan.	A full spectrum of cases—a parvovirus puppy, a blocked cat, an oxygen-dependent budgie, a dehydrated bearded dragon, a car-struck kitten, and surgical foreign-body and laceration management—interleaved with hard owner conversations about cost and end-of-life decisions while teaching and caring for his techs.
Carl (Pilot)	Chicago-based commercial airline pilot in his career-progression phase flying the ORD–LAX route; doting father, deep-dish-pizza regular, and casual flight-prep-plus-food-and-music conversationalist.	A single early-October trip: flying an ORD–LAX leg, completing a multi-day captain-upgrade training course in Los Angeles, earning the training-completion certificate, and returning to Chicago to celebrate.
Claire (Cat Cafe)	Runs the Cat Loft, a small cat-cafe and kitten-rescue space; warm, practical, and patient, handling kitten care, pregnant-stray intake, vet visits, adoption matchmaking, and social media, baking cat-friendly catnip treats for adoptive families.	Scaling up the rescue: a community TNR effort, raising and placing several kitten litters (including one from a pregnant stray she helps deliver), weathering a sick-kitten scare, and growing the cafe’s social-media reach, all with a hands-on caretaking approach.
Ethan (Violinist)	Career violinist and visiting orchestra coach at a Seattle arts high school; gentle, patient, and documentation-minded, deliberate about his energy; a devoted Seattle ramen hunter who hosts Studio Ghibli nights with students and dotes on his niece Lily.	A visiting-coach residency: preparing rehearsals, mediating a student conflict, and troubleshooting a stage-lighting failure right before the concert, closing with a framed certificate at the post-concert dinner, all in his gentle, student-first style.
Liam (Woodworking Apprentice)	Self-taught woodworking apprentice climbing a ladder of projects from chisel sharpening and a small box up to a dining table and wardrobe; humble, curious, and detail-oriented, asking practical “any tips?” questions and studying American hardwoods on the side.	Climbing a deliberate apprenticeship ladder from a first box and bird carving through bookshelf and birdhouse builds, graduating to a solid-wood dining table and a full wardrobe, with woodworking-class study material complementing the bench work.
Lily (Cooking Newbie)	Enthusiastic first-time cook buying her first knife, board, and wok and working up from basic stir-fries to omelets, sauces, pasta, and a quiche; curious and process-oriented, sharing photos and asking “did I do it right?”, loving the small wins of home cooking.	Moving from a first market trip and basic stir-fry through knife skills, sauces, omelets, smoothie bowls, pasta, and a quiche, building a starter repertoire and the confidence to take on a community-cooking class studying nutrition and recipe guides.
Lin Zeyu (CS Junior)	Computer Science junior juggling math and systems coursework with backend and full-stack project work (SQL, Spring Boot, Vue, Docker, Git, Redis); technical, direct, and tenacious, levelling up toward an industry SDE role on late-night coding sessions and cheap fried rice.	A full semester from foundational coursework into a capstone: slow SQL queries, Spring Boot startup failures, Vue bugs, CORS issues, and full backend–frontend integration, culminating in a deployed remote-server stack plus CS-paper self-study.
Lin (Bike Repair)	Runs a small, busy bike-repair shop in a tourist town; hands-on wear diagnostics, brake and bottom-bracket service, e-bike checks, vintage restoration, and trainee mentoring; practical and relationship-first, enjoys weekend group rides, and keeps studying Shimano manuals.	A packed month at the start of tourist season: managing a repair surge while professionalising the shop with a safety-disclosure script, trainee training, a beginner workshop, restocking, and a vintage-bike restoration, keeping pricing honest and customers front and centre.
Lina (Family)	Busy mother of two (Emma on piano and presentation nerves, Brandon on calculus and track) managing her own hyperthyroidism and running the household; warm, careful, a touch anxious, putting family wellness first and always trying to cook a little better.	A normal-life family stretch: supporting both kids through their worries, handling a hyperthyroidism clinic visit, troubleshooting a washing-machine overflow, smoothing over parent-meeting tensions, and reining in grocery costs without anyone feeling squeezed.
Marcus (Travel)	Unhurried solo traveller on a multi-country trip (Japan, Vietnam, Thailand, France, Spain); experiences-over-things style and a keen amateur photographer writing for a personal travel blog, hunting unusual local snacks and crafts for his niece Lily.	A deliberate Asia-to-Europe arc: Osaka ryokans and temples, Hoi An street food, Bangkok and Chiang Mai markets, Montmartre wine-and-cheese, and Barcelona’s Gaudí architecture, photographing each stop for his blog and gathering snacks and crafts for Lily.
Mathieu (Reservoir Engineer)	Reservoir engineer at TotalEnergies in Pau, France; well-log interpretation, 3D Petrel geomodels, Eclipse simulation, history matching, and waterflood optimisation, with a growing CCS and energy-transition interest; surfs the Atlantic coast on weekends.	His structured first months at TotalEnergies: joining the asset team, learning integrated log-to-model interpretation, building first geomodels and simulations, and presenting field-development recommendations, later digging into the company’s financial report and energy-transition strategy.
Mina (Botany Student)	Undergraduate botany student building a daily plant-identification practice (leaf arrangement, venation, stress diagnosis, herbarium, invasive-species project); careful and evidence-led, leaning on iNaturalist but cross-checking with textbooks and TAs, and loving weekend nature walks.	A semester building a botany toolkit through campus identification, venation and trichome documentation, an invasive-species visual guide, and garden visits, then a deeper-reading phase on plant phylogenomics, marking her shift toward graduate-level research.
Noah (Barista Apprentice)	Barista apprentice at a specialty coffee shop methodically working through craft fundamentals (pour-over, espresso dial-in, grinder settings, microfoam, cold brew, TDS); technical and detail-led, geeking out over single-origin beans and supplementing with barista training manuals.	A structured apprenticeship arc from pour-over and espresso basics through cold brew, microfoam, single-origin dial-in, water quality, and batch brewing, culminating in a signature-blend capstone, followed by deeper study of specialty-coffee training material.
Oliver (PhD CS)	CS PhD student working on Transformer architectures, RAG systems, and vector retrieval (FAISS, Milvus); mixes deep technical debugging with academic rhythms of ablations, drafts, and LaTeX revisions; a devoted campus-coffee-shop regular powering late-night writing pushes.	A complete PhD research arc: Transformer reading, HuggingFace and training-pipeline debugging, building and tuning RAG and vector-retrieval systems on Milvus, running an ablation, and revising a paper draft, wrapping with foundational-paper study, often from his favourite coffee shop.
Wei (Courier)	Food-delivery courier in a tier-2 Chinese city working surge-driven app-dispatched shifts; route optimisation across modes, pickup pacing, and turnstile navigation; hardworking, resourceful, and budget-conscious, getting by on cheap eats and a rented single room, and studying labour-conditions papers.	An intensive shift cluster running full courier rotations through rainy-brake scares, e-bike-vs-motorcycle cost decisions, office-turnstile gauntlets, marathon detours, and late-night surges, while managing rent, meals, and rest, later studying ergonomic and platform-governance papers.

E Prompt Templates

We list the prompt templates in the order they are invoked while constructing the dataset.

Timeline construction. Generation begins from the persona and core event. We first prompt the LLM with the *Timeline Generation* template (Figure 13) to produce the core-event line of N chronological events, each carrying a description, a visual-grounded user query, an absolute date, and an image-search keyword. We then call the *Distraction Timeline Generation* template (Figure 14) to add M off-topic side events drawn from orthogonal domains, which are interleaved into the core line by date. The merged timeline is passed through one round of the *Timeline Self-Check* template (Figure 15), which audits persona violations, core-event drift, and internal contradictions and flags any offending events for repair. The audited timeline is then released to a human reviewer for a final spot-check before it enters dialogue synthesis.

Dialogue synthesis. With the audited timeline in hand, we generate the conversation by walking an event sliding window: each round feeds the current event (and, after the first round, the immediately preceding one) into the *Thematic User Simulator* template (Figure 16), which produces the user’s motivated query over the event’s attached images. The simulator is conditioned on the running conversation summaries so that follow-up queries can revisit earlier context. On the *first* round the window contains only the opening event and there is no prior summary, so the simulator emits a single scene-setting question that introduces the persona’s situation rather than a follow-up. When the shared evidence for an event is a *document* rather than a photo, the same simulator role is used with two differences: the user is shown a batch of freshly rendered PDF pages instead of images and must ask a question that quotes a concrete number, label, or term from those pages; and the assistant answers using only the visible pages, saying so when the answer is not present. The user query is then passed to the *Thematic Assistant* template (Figure 17), which replies over the same attached evidence. After each round, the exchange is condensed by the *Round Summary* template (Figure 20) and appended to the running memory that conditions the next round.

Dialogue quality control. Once all rounds are generated, the dialogue is audited in two passes. The *Dialogue Self-Check (Per-Chunk)* template

(Figure 18) scans a contiguous slice of rounds for persona violations, core-event drift, hallucinations, and within-chunk contradictions; the *Dialogue Self-Check (Cross-Chunk)* template (Figure 19) then compares distant rounds to catch contradictions and invariant failures that no single chunk reveals. Flagged rounds are repaired, and the *Round Summary* template (Figure 20) provides the per-round summaries used both for cross-round memory during generation and as the compact dialogue overview consumed by the cross-chunk pass.

Retrieval-augmented generation. For all memory-based systems, at inference time the prompt is composed from the retrieved information and any attached images or rendered PDF pages. The base answering template (top of Figure 22) requests a short, direct answer and is shared by all question types. For find-matching (FM) questions, where the gold answer is an image identifier rather than free text, the template is extended with the additional instruction shown below the rule, which constrains the model to choose among the candidate file names actually attached from the retrieved rounds, preventing it from inventing or emitting the placeholder `img_<number>`.

F Configuration Reference

All models are evaluated with deterministic generation whenever the endpoint supports it. The default settings are temperature 0.0, top- $p = 1.0$, and a maximum of 100 LLM calls per turn.

All target models are exposed to the benchmark through the same OpenAI-compatible function-calling scaffold. Local vLLM services use tensor parallelism with two to four A100 PCIe 80GB GPUs per model, bfloat16 inference, a 32,768-token maximum context unless explicitly overridden, prefix caching, chunked prefill, up to 32 concurrent sequences, and a maximum batched-token budget of 65,536. For local vLLM models, model-family-specific parsers and chat templates are used only inside the serving layer to translate model-native outputs into this common interface: Qwen uses the Qwen tool-call parser with thinking disabled, while Gemma and GLM use Gemma- and GLM-compatible chat templates and tool-call parsers.

Timeline Generation

You are generating a timeline of events for a synthetic user's life. Return ONLY a JSON array. No markdown. No trailing commentary.

Persona: <persona>

Core event: <core_event>

Hard requirements:

- Output EXACTLY <N> event objects (no more, no fewer).
- Every string field below must be NON-EMPTY. Never use "" or omit a key.
- Chronological Event_Time across the story.

Each event MUST be an object with these keys in order: (1) Event_Index (integer 1..N matching array position); (2) Event_Description (one third-person sentence); (3) Query_Description (one sentence using the protagonist name: "[Name] shares photos/screenshots of [concrete visuals] and asks [one specific question]"); (4) Event_Time (YYYY-MM-DD); (5) Keyword (a 2–6 word image-search phrase).

Figure 13: Prompt template for timeline generation. The event count <N> is set by the configuration.

Distractor Timeline Generation

You are generating DISTRACTOR life events for the same synthetic user, to be interleaved with their main story. These events must look like ordinary moments in this person's life but must NOT advance, mirror, or reference the core event arc. Return ONLY a JSON array.

Persona: <persona>

Core event (DO NOT touch these themes): <core_event>

Existing main-line events (date range only): <mainline_context>

Hard requirements:

- Output EXACTLY <M> event objects.
- Each event is a believable, mundane moment ORTHOGONAL to the core event (food / errands / hobbies / minor health / weather / family chat / commute / app glitch).
- Event_Time values fall WITHIN the main-line date range and interleave plausibly (do not bunch on one date).

Keys per object: Event_Description, Query_Description (off-topic visual + unrelated question), Event_Time, Keyword (off-topic).

Figure 14: Prompt template for distractor (side-event) generation.

Timeline Self-Check

You are auditing a generated timeline against the persona and the core event. Return ONLY a JSON object.

Persona: <persona> **Core event:** <core_event>

Timeline (JSON array of events): <timeline>

Check three things and report each finding with the offending Event_Index values:

- **persona_violations** — events contradicting the persona (wrong name, profession, impossible behaviour).
- **core_event_violations** — main-line events that fail to express or contradict the core-event arc (events marked off-topic are exempt).
- **contradictions** — pairs of events that contradict each other (impossible date order, mutually exclusive states, name drift).

Output a JSON object with the three arrays above plus "ok" (true iff all are empty).

Figure 15: Prompt template for the timeline self-check audit.

Thematic User Simulator

You are the USER in a roleplay.

Persona: <persona>

Relevant event(s) (JSON): <events>

Conversation summaries so far (JSON): <summaries>

Attached images: <image_desc>

Write TWO consecutive user messages that progress the conversation. Return ONLY a JSON array of TWO objects, each with keys `speaker` (= "User"), `timestamp`, `clean_text`, `image_description`.

Figure 16: Prompt template for the thematic user simulator (follow-up turn).

Thematic Assistant

You are the ASSISTANT.

Users said (JSON array of 2 user messages): <user_messages>

Attached images: <image_desc>

Reply to each user message in order with TWO assistant messages. Return ONLY a JSON array of TWO objects, each with keys `speaker` (= "Agent") and `clean_text`.

Figure 17: Prompt template for the thematic assistant.

Dialogue Self-Check (Per-Chunk)

You are auditing a contiguous slice of a generated dialogue against the persona and the core event. Return ONLY a JSON object.

Persona: <persona> **Core event:** <core_event>

Timeline overview: <timeline>

Dialogue rounds in this chunk (each with a round id like D1:7): <chunk>

Audit four categories, each reporting the offending round id:

- **persona_violations** — a turn contradicts the persona.
- **core_event_violations** — main-line rounds that fail to advance the core-event arc (off-topic rounds are fine).
- **hallucinations** — the assistant invents entities/data/numbers never grounded by persona, core event, timeline, or earlier rounds.
- **internal_contradictions** — within this chunk, two rounds that contradict each other.

Output the four arrays plus "ok" (true iff all are empty).

Figure 18: Prompt template for the per-chunk dialogue self-check.

Dialogue Self-Check (Cross-Chunk)

You are doing a CROSS-CHUNK pass over a long generated dialogue, looking for contradictions that span chunks (not visible inside any single chunk). Return ONLY a JSON object.

Persona: <persona> **Core event:** <core_event>

Per-chunk findings already collected: <chunk_findings>

Dialogue overview (each round's user line summarised with its id): <overview>

Flag problems where one round contradicts a much earlier or later round, or where a persona / core-event invariant fails when read end-to-end. Do NOT repeat per-chunk findings; report only NEW issues requiring distant comparison. Output `cross_chunk_contradictions`, `global_persona_violations`, `global_core_event_violations`, and "ok".

Figure 19: Prompt template for the cross-chunk dialogue self-check.

Round Summary

Summarize the following USER+ASSISTANT exchange in 1–2 short sentences. Stay faithful to what was actually said; do not add facts or numbers not in the exchange. Paraphrase at the level of topics and numbers; do not rely on file names.

Exchange: <exchange>

Return ONLY a flat JSON object with exactly one string field "summary".

Figure 20: Prompt template for per-round summarisation (cross-round memory).

LLM-as-a-Judge

You are an impartial judge evaluating the memory capabilities of an AI assistant on a question-answering task. Compare the Assistant's Answer against the Ground Truth and assign a score of 0, 0.25, 0.5, 0.75, or 1.

Question: <question> **Gold Answer:** <gold> **Model Answer:** <model_answer>

Scoring Rubric

- **Score 0 (Incorrect / Miss):** the answer contradicts the Ground Truth; for Yes/No questions the polarity is wrong; for open-ended questions it is factually wrong or hallucinated; or it fails to provide the required information.
- **Score 0.25 (Poor / Tangential):** touches the topic but misses the core entity or key value; mixes minor correct details with significant hallucinations; or is uselessly vague (e.g., “a dog” instead of “a golden retriever”).
- **Score 0.5 (Partial / Vague / Excessive):** technically correct but incomplete or unconfident; captures the main entity but misses required supporting details; or is over-informative. For Yes/No, the polarity is correct but reasoning is flawed or uncertain.
- **Score 0.75 (Good / Minor Imperfection):** largely accurate and confident, missing only minor details, or correct but with unnecessary filler that reduces precision.
- **Score 1 (Correct / Exact):** accurate, precise, and confident; for Yes/No the polarity matches exactly; for open-ended it contains all core information and necessary details without hallucinations.

Reply with ONLY the score digit. No other text.

Figure 21: Prompt template for the LLM-as-a-Judge metric (five-level rubric).

Retrieval-Augmented Generation

You are answering a memory-QA question. Use ONLY the information below (and any attached images / rendered PDF pages).

=== INFORMATION ===

<information>

=== END ===

Question: <question>

Give a short, direct answer (a few words or a brief phrase). Do not add explanations or repeat the question.

Additional instruction for find-matching (FM) questions:

The candidate images attached below come from the retrieved rounds. If the question asks for image file name(s), reply with one to three matching file names from this list (comma-separated): <candidate_filenames>. Do not invent file names that are not in this list. Do not respond with the placeholder img_<number>.

Figure 22: Answering prompt used by M³Proctor. The base template (top) is shared by all question types; the part below the rule is appended only for find-matching-image (FM) questions.